

Topological Regime Detection in Financial Markets via Quantum Geometric Machine Learning

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Abstract

Classical regime detection methods operate in flat Euclidean feature spaces, relying on distributional assumptions that are violated during the very transitions they seek to detect. We construct a learned Riemannian manifold for financial markets by lifting time series into a projective Hilbert space $\mathbb{CP}^{n-1}$ via an error Hamiltonian and extracting the induced geometric structure: quantum metric tensor, Berry curvature, spectral gap, and quantum Fisher information (QFI) matrix. This geometry induces a family of scalar observables—Berry phase rate, QFI pseudo-determinant, rolling Chern numbers, multi-lag fidelity, and metric condition number—that summarize the local and global structure of the market manifold at each point in time. In a head-to-head comparison of 16 methods across 12 historical crises (2007–2023), we show that: **(i)** Optuna-optimized fusion of the three top geometric observables achieves median Cohen’s $d = 1.67$ across all crises—48% above the supervised Random Forest baseline ($d = 1.13$) despite requiring no labeled training data; **(ii)** under a strictly causal protocol that prevents lookahead bias, individual geometric observables achieve $d > 2.0$ (QFI $d = 2.30$, multi-lag fidelity $d = 2.29$); **(iii)** the geometric framework generalizes across asset classes: all three methods maintain $d > 1.0$ on five broad-market ETFs (SPY, QQQ, IWM, EFA, DIA) with Friedman $\chi^2 = 35.0$, $p < 0.0001$; and **(iv)** the geometric signal is complementary to supervised detection—on novel crises (SVB, COVID), QCML achieves $d > 1.7$ where Random Forest scores $d < 0.5$, while a hybrid ensemble nearly doubles out-of-sample d relative to RF alone ($d = 0.60$ vs. $d = 0.31$). These findings establish the learned market manifold as a principled geometric framework whose topological and curvature properties encode regime structure, with greatest value precisely where supervised approaches are weakest.

Keywords: regime detection, quantum geometry, Chern number, Berry curvature, quantum metric, topological invariants, financial crises, quantum Fisher information.

1 Introduction

Detecting regime transitions is a central problem in quantitative finance. Market dynamics alternate between qualitatively different states—low-volatility trends, high-volatility drawdowns, liquidity crises—and the ability to identify transitions in real time has implications for risk management, portfolio construction, and hedging.

Classical approaches include Hidden Markov Models (HMMs) [1], cumulative-sum (CUSUM) statistics [2], rolling volatility breakpoints, and supervised classifiers such as Random Forests. While effective in specific settings, these methods share a common limitation: they operate entirely within

a Euclidean feature space and rely on distributional assumptions (Gaussian mixtures, stationarity within regimes) that are frequently violated during the very transitions they seek to detect.

Our central thesis is that financial regimes correspond to distinct topological phases of a learned market manifold, and that regime transitions manifest as geometric singularities—curvature spikes, spectral gap closures, and changes in topological charge—of this manifold.

In this paper we construct a learned Riemannian manifold for financial markets and show that regime detection emerges as a natural consequence of its geometry. Inspired by the Quantum Cognition Machine Learning (QCML) framework [3], we *lift* the feature space $\mathbb{R}^d$ into a projective Hilbert space $\mathbb{CP}^{n-1}$ via a family of Hermitian operators. The ground state $|\psi(x)\rangle$ of an error Hamiltonian $H(x)$ defines a smooth map whose differential geometry inherits topological structure from quantum mechanics. Specifically:

1. The *quantum metric tensor* g_{ab} (the real part of the Fubini–Study metric pulled back to $\mathbb{R}^d$) endows the data manifold with a natural Riemannian structure, replacing the ad hoc Euclidean metric.
2. The *Berry curvature* F_{ab} (the antisymmetric, imaginary part) [15] gives rise to topological invariants—*Chern numbers*—that are integers for closed manifolds and therefore intrinsically robust to continuous perturbations (noise).
3. Discontinuities in the rolling Chern number distinguish *topological* regime transitions ($\Delta C \neq 0$) from extreme events within the same regime ($\Delta C = 0$), a distinction inaccessible to purely statistical detectors.

Contributions. Our contributions form a three-layer construction with empirical validation:

1. **The geometry.** We construct the full Riemannian and topological geometry of a financial data manifold via the QCML error Hamiltonian, including the quantum metric tensor, Berry curvature, Christoffel symbols, Ricci scalar, spectral gap, and quantum distances (Section 3).
2. **Geometric dynamics.** We formulate a stochastic differential equation whose diffusion tensor is derived from the inverse quantum metric, producing dynamics that respect the learned geometry (Section 4).
3. **Geometric observables and their regime-detection consequences.** The geometry induces a family of scalar observables—Berry phase rate, QFI pseudo-determinant, Chern numbers, multi-lag fidelity, metric condition number, and five additional indicators (Section 5). We show that these observables change sharply around known regime transitions across 12 historical crises on five broad-market ETFs. Optimized fusion of the top three observables achieves median $d = 1.67$, exceeding the supervised Random Forest baseline ($d = 1.13$) by 48%, and individual methods reach $d > 2.0$ under strictly causal protocols (Section 7).

2 Background and Related Work

2.1 Classical Regime Detection

The regime-switching framework originates with Hamilton’s Markov-switching autoregression [1], in which a latent discrete state governs the parameters of an observed process. The two-state Gaussian HMM remains a workhorse, estimating $P(\text{high-vol state} | r_t)$ via the Baum–Welch algorithm. CUSUM statistics [2] provide a nonparametric alternative, accumulating deviations $S_t =$

$\max(0, S_{t-1} + |r_t| - k)$ and flagging $S_t > h$ as a changepoint. Rolling volatility z-scores are perhaps the simplest approach: one computes $z_t = (\sigma_t^{\text{short}} - \mu_t^{\text{long}})/\sigma_t^{\text{long}}$ and applies a threshold. Supervised classifiers (Random Forest, gradient-boosted trees) can achieve high in-sample accuracy but require labeled data and are prone to overfitting on rare events [4].

Bayesian online changepoint detection. Adams and MacKay [16] introduced Bayesian Online Changepoint Detection (BOCPD), which maintains a posterior distribution over the *run length*—the time since the last changepoint—and updates it recursively as new observations arrive. Extensions to multivariate settings and non-exponential-family likelihoods have been explored by Knoblauch and Damoulas [17]. While BOCPD provides principled uncertainty quantification over changepoint timing, it remains a distributional method operating in flat Euclidean space.

All of these methods treat features as elements of a flat Euclidean space. Our approach differs by constructing a learned Riemannian manifold whose curvature and topology encode structural information about the market state—and whose geometric observables, as we show, change sharply at regime boundaries.

2.2 Quantum-Inspired Machine Learning

The QCML framework [3] embeds classical data into quantum mechanical structures. Each data point $x \in \mathbb{R}^d$ is associated with a Hamiltonian $H(x)$ whose ground state $|\psi(x)\rangle$ serves as a nonlinear feature map $\mathbb{R}^d \rightarrow \mathbb{CP}^{n-1}$. This is conceptually analogous to kernel methods—the Fubini–Study metric induces a positive-definite kernel $K(x, y) = |\langle \psi(x) | \psi(y) \rangle|^2$ —but additionally provides geometric and topological structure absent from standard kernel machines.

2.3 Topological Data Analysis in Finance

Topological data analysis (TDA) applies algebraic topology to point-cloud data. Gidea and Katz [18] used persistent homology on sliding windows of financial time series, computing Betti numbers β_0 (connected components) and β_1 (loops) of Vietoris–Rips complexes to detect early warning signals of market crashes. Gidea [19] extended this to cross-sectional equity data, showing that changes in topological features precede major drawdowns. Persistent landscapes [20] and persistence images [21] provide stable vectorizations of persistence diagrams suitable for downstream machine learning.

Our approach differs from TDA in several respects. First, TDA extracts topology from point-cloud geometry (pairwise distances), while we extract it from a *learned* quantum-mechanical structure (Berry curvature and Chern numbers). Second, TDA typically computes global topological invariants over a sliding window, whereas our Chern numbers are local quantities defined at each point on the parameter manifold. Third, our framework provides a richer geometric structure (metric tensor, curvature, spectral gap) beyond topology alone. We compare directly to a TDA baseline in Section 7.

2.4 Deep Learning Approaches to Regime Detection

Deep learning methods have been applied to regime detection with mixed results. Recurrent architectures such as Long Short-Term Memory (LSTM) networks [22] can model temporal dependencies in financial time series, while Temporal Convolutional Networks (TCNs) [23] offer parallelizable alternatives with dilated causal convolutions. Autoencoder-based anomaly detection [24] identifies regime transitions as reconstruction errors. Attention mechanisms [25] have been applied to changepoint detection, learning which temporal features are most informative for regime classification.

These deep learning approaches are fundamentally supervised: they require labeled regime data for training and tend to overfit on the small number of historical crises available. Our geometric observables are unsupervised—they detect regime transitions as intrinsic geometric singularities without labeled training data—making them particularly valuable for detecting structurally novel crises.

2.5 Geometric Methods in Finance

Information geometry [7] studies the differential geometry of statistical models. The Fisher information metric on the space of probability distributions has been applied to financial time series analysis [8], where geodesic distances between estimated distributions measure regime dissimilarity. Our quantum metric g_{ab} is the natural generalization of the Fisher metric to the projective Hilbert space setting: when the parametric family $|\psi(x)\rangle$ is interpreted as a statistical model (via Born’s rule), g_{ab} reduces to $\frac{1}{4}$ times the quantum Fisher information matrix [6].

Rough paths and path signatures. The theory of rough paths [26] provides a coordinate-free description of irregular paths, and the *path signature*—the iterated integrals of a path—serves as a universal nonlinear feature map. Chevyrev and Kormilitzin [27] applied signatures to sequential data classification, and Lyons et al. [28] demonstrated their effectiveness for financial time series analysis. The path signature captures order-dependent information (lead-lag relationships) that Euclidean features miss. Our approach is complementary: while signatures characterize the path in a feature space, our geometry characterizes the *manifold structure* induced by projecting into Hilbert space.

2.6 Quantum Finance

Quantum-inspired approaches to finance extend beyond the QCML framework. Baaquie [29] formulated option pricing using path integrals over field configurations, recovering Black–Scholes in appropriate limits. Haven and Khrennikov [30] developed quantum probability models for decision-making under Knightian uncertainty, showing that the non-commutativity of quantum observables naturally models order effects and conjunction fallacies in behavioral finance. Piotrowski and Śladkowski [31] formulated market games using quantum strategies.

Our work differs from quantum finance proper in that we do not posit quantum mechanics as a fundamental model of markets; rather, we use the *mathematical machinery* of quantum mechanics (Hilbert spaces, Hermitian operators, Berry phases) as a structured nonlinear embedding that provides geometric and topological tools unavailable in classical feature spaces.

2.7 SDEs in Quantitative Finance

Stochastic differential equations are the standard modeling framework in quantitative finance, from Black–Scholes [9] to the Heston stochastic volatility model [10]. The geometric SDE we construct in Section 4 differs from these classical models in that its diffusion tensor is not assumed *a priori* but is *learned* from the data via the quantum metric.

3 QCML-Induced Market Geometry

The central mathematical object of this paper is a Riemannian manifold learned from financial data. We construct it by lifting the feature space into a projective Hilbert space via an error Hamil-

tonian, and then extracting the induced geometric structure—quantum metric, Berry curvature, spectral gap, and higher invariants—as properties of the market state space. All constructions are implemented in the `QCMLGeometry` class; we include code-level references for reproducibility.

3.1 Error Hamiltonian and Quasi-Coherent States

Let $\{A_k\}_{k=1}^d$ be a family of $n \times n$ Hermitian operators (observables) learned from data. Given a data point $x \in \mathbb{R}^d$, we define the *error Hamiltonian*:

$$H(x) = \frac{1}{2} \sum_{k=1}^d (A_k - x_k I)^2, \quad (1)$$

where I is the $n \times n$ identity matrix. The Hamiltonian $H(x)$ is Hermitian and positive semi-definite by construction. Intuitively, $H(x)$ measures the collective “error” between the observables $\{A_k\}$ and the target eigenvalues $\{x_k\}$: a data point x that is well-represented by the operators yields a small ground state energy.

Definition 1 (Quasi-coherent state). *The quasi-coherent state at x is the ground state of $H(x)$:*

$$|\psi(x)\rangle = \arg \min_{|\phi\rangle, \|\phi\|=1} \langle \phi | H(x) | \phi \rangle. \quad (2)$$

The map $x \mapsto |\psi(x)\rangle$ is a smooth embedding $\mathbb{R}^d \hookrightarrow \mathbb{CP}^{n-1}$ wherever the ground state is non-degenerate (i.e., the spectral gap $\Delta(x) > 0$). This map lifts each market observation into the projective Hilbert space, endowing the data manifold with quantum-mechanical geometry.

Theorem 2 (Smoothness of the QCML embedding). *Let $\{A_k\}_{k=1}^d$ be Hermitian operators and let $H(x)$ be the error Hamiltonian (1). If the spectral gap $\Delta(x) = E_1(x) - E_0(x) > 0$ for all x in an open set $U \subset \mathbb{R}^d$, then:*

1. *The ground state map $x \mapsto |\psi(x)\rangle \in \mathbb{CP}^{n-1}$ is C^∞ on U .*
2. *The quantum metric tensor $g_{ab}(x)$ and Berry curvature $F_{ab}(x)$ are C^∞ on U .*
3. *All geometric observables derived from g_{ab} and F_{ab} (Christoffel symbols, Ricci scalar, Chern numbers) are C^∞ on U .*

The proof follows from the implicit function theorem applied to the eigenvalue equation $H(x)|\psi(x)\rangle = E_0(x)|\psi(x)\rangle$: the non-degeneracy condition $\Delta > 0$ ensures that the ground state depends smoothly on parameters. See Appendix B for details.

Operator learning. The operators $\{A_k\}$ are constructed from the data covariance structure. In the *PCA-inspired* method, we compute the principal components of the data matrix X , then scale Pauli-basis operators by the corresponding eigenvalues: $A_k = \sqrt{\lambda_k} \sigma_k$, where σ_k are tensor products of Pauli matrices (for $n = 4$, the two-qubit basis $\{I \otimes I, I \otimes X, \dots, Z \otimes Z\}$). Alternative methods include random Hermitian matrices and pure Pauli bases.

3.2 Quantum Metric Tensor

The pullback of the Fubini–Study metric [14] to $\mathbb{R}^d$ defines the *quantum metric tensor*:

$$g_{ab}(x) = \text{Re} \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle, \quad (3)$$

where $\partial_a \equiv \partial/\partial x^a$. This is a real, symmetric, positive semi-definite tensor that generalizes the Fisher information metric to the Hilbert space setting. It measures the sensitivity of the quantum state to infinitesimal changes in the data coordinates: large g_{ab} indicates that nearby data points map to distant states in $\mathbb{CP}^{n-1}$.

Partial derivatives are computed numerically via central finite differences:

$$|\partial_a \psi(x)\rangle \approx \frac{|\psi(x + \epsilon e_a)\rangle - |\psi(x - \epsilon e_a)\rangle}{2\epsilon}, \quad (4)$$

with $\epsilon = 10^{-5}$.

3.3 Berry Curvature

The antisymmetric part of the quantum geometric tensor defines the *Berry curvature* [11, 13]:

$$F_{ab}(x) = -2 \text{Im} \langle \partial_a \psi | \partial_b \psi \rangle. \quad (5)$$

This is a closed 2-form on $\mathbb{R}^d$ that encodes topological information. While the metric tensor g_{ab} is sensitive to parameterization, the integral of F_{ab} over closed surfaces yields topological invariants (Chern numbers) that are integers [12] and therefore robust to smooth deformations.

For two-dimensional subspaces, we also implement a *plaquette* (Wilson loop) method that is more numerically stable for computing Chern numbers. The Berry phase around a plaquette with corners $x_{00}, x_{10}, x_{11}, x_{01}$ is:

$$\gamma = \text{Im} \log(\langle \psi_{00} | \psi_{10} \rangle \langle \psi_{10} | \psi_{11} \rangle \langle \psi_{11} | \psi_{01} \rangle \langle \psi_{01} | \psi_{00} \rangle), \quad (6)$$

where the argument of the logarithm is the Wilson loop operator. The Berry curvature is then $F_{ab} \approx \gamma/\epsilon^2$.

Theorem 3 (Chern number quantization). *Let $S \subset \mathbb{R}^d$ be a closed, oriented 2-surface in parameter space over which the ground state is non-degenerate ($\Delta > 0$ on S). Then the first Chern number*

$$C_1 = \frac{1}{2\pi} \oint_S F_{ab} dx^a \wedge dx^b \in \mathbb{Z} \quad (7)$$

is an integer. In particular, C_1 is invariant under smooth deformations of the Hamiltonian parameters that preserve $\Delta > 0$.

This is a consequence of the Chern–Weil theorem applied to the $U(1)$ line bundle over S defined by the ground state $|\psi(x)\rangle$; see Appendix B for the proof. The quantization of C_1 is the mathematical basis for our claim that Chern-number-based regime detection is *topologically protected*: small perturbations (noise) cannot change an integer.

3.4 Spectral Gap

The *spectral gap* at x is the difference between the two lowest eigenvalues of the error Hamiltonian:

$$\Delta(x) = E_1(x) - E_0(x). \quad (8)$$

A large gap indicates a well-defined, non-degenerate ground state and hence stable geometry. A closing gap signals the approach to a *quantum critical point*—in our financial analogy, a regime transition [5]. The spectral gap also controls the accuracy of the adiabatic approximation: when $\Delta \rightarrow 0$, the system is susceptible to non-adiabatic transitions between eigenstates.

Theorem 4 (Curvature–gap bound). *For any pair of indices (a, b) with $a \neq b$, the Berry curvature is bounded by the spectral gap:*

$$|F_{ab}(x)| \leq \frac{K}{\Delta(x)^2}, \quad (9)$$

where $K = 2 \sum_{n \neq 0} |\langle n | \partial_a H | 0 \rangle| |\langle n | \partial_b H | 0 \rangle|$ depends on the Hamiltonian derivatives but not on the gap.

This bound follows from first-order perturbation theory for the Berry connection (see Appendix B). It formalizes the mechanism underlying curvature-based regime detection: as the spectral gap $\Delta \rightarrow 0$ at a phase boundary, the Berry curvature can diverge as Δ^{-2} , producing the sharp spikes that our detectors identify. The converse also holds qualitatively: in stable regimes where Δ is large, curvature fluctuations are suppressed.

3.5 Quantum Fisher Information Susceptibility

The *QFI susceptibility* is defined as the trace of the quantum metric tensor:

$$\chi(x) = \text{tr}(g_{ab}(x)) = \sum_a g_{aa}(x). \quad (10)$$

This scalar quantity equals $\frac{1}{4}$ times the total quantum Fisher information and measures the aggregate sensitivity of the quantum state to parameter changes. Near regime transitions, the market state becomes maximally sensitive to perturbations, causing χ to spike—analogous to divergent susceptibility at quantum critical points [6].

We further define the *QFI pseudo-determinant* as

$$\det^+(g) = \prod_{\lambda_i > \epsilon} \lambda_i, \quad (11)$$

where $\{\lambda_i\}$ are the eigenvalues of g_{ab} and $\epsilon = 10^{-10}$ filters near-zero eigenvalues for numerical stability. The pseudo-determinant measures the volume element of the non-degenerate subspace of the data manifold: a collapsing pseudo-determinant indicates that the effective dimensionality of the market state is decreasing (approaching a phase boundary), while a diverging pseudo-determinant indicates rapid expansion of the accessible state space.

Proposition 5 (QFI volume interpretation). *The pseudo-determinant $\det^+(g)$ equals the squared Riemannian volume element of the non-degenerate subspace of the data manifold. Specifically, if g_{ab} has r positive eigenvalues $\lambda_1 \geq \dots \geq \lambda_r > 0$, then*

$$\det^+(g) = \prod_{i=1}^r \lambda_i = (\text{vol}_r)^2, \quad (12)$$

where vol_r is the r -dimensional volume form induced by the quantum metric restricted to the non-degenerate subspace. Thus:

1. $\det^+(g) \rightarrow 0$ indicates dimensional collapse of the accessible state space (approaching a phase boundary).
2. Large $\det^+(g)$ indicates the market state is exploring a high-dimensional region of quantum state space (stable, multi-factor regime).

The proof is a direct consequence of the relationship between the determinant of a metric tensor and the Riemannian volume element; see Appendix B.

3.6 Quantum Distances

The *Fubini–Study distance* between two data points is:

$$d_{\text{FS}}(x_1, x_2) = \arccos(|\langle \psi(x_1) | \psi(x_2) \rangle|). \quad (13)$$

This gives a natural distance on the data manifold that respects the quantum geometry: data points in the same regime have high fidelity $|\langle \psi(x_1) | \psi(x_2) \rangle|^2 \approx 1$ and small d_{FS} , while points in different regimes have low fidelity and large d_{FS} .

3.7 Higher Geometry: Christoffel Symbols and Ricci Scalar

From the quantum metric g_{ab} , we compute the Levi-Civita connection (Christoffel symbols):

$$\Gamma_{\mu\nu}^\sigma(x) = \frac{1}{2} g^{\sigma\rho} (\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}), \quad (14)$$

and the Ricci scalar curvature:

$$R(x) = g^{\mu\nu} R_{\mu\nu}, \quad (15)$$

where $R_{\mu\nu} = R^\sigma_{\mu\sigma\nu}$ is the Ricci tensor obtained by contracting the Riemann curvature tensor. We employ a *hierarchical finite difference* scheme with three levels of step size ($\epsilon_{\text{metric}} = 10^{-5}$, $\epsilon_{\text{Christoffel}} = 10^{-4}$, $\epsilon_{\text{Ricci}} = 10^{-3}$) to avoid differentiating numerical noise.

Large $|R|$ indicates regions of high intrinsic curvature on the data manifold, corresponding to nonlinear, unstable dynamics.

The geometric objects defined above—metric tensor g_{ab} , Berry curvature F_{ab} , spectral gap Δ , Ricci scalar R , QFI matrix $\mathcal{F}$ —constitute a complete differential-geometric description of the learned market manifold. In subsequent sections, we show that the dynamics (Section 4) and observables (Section 5) derived from this geometry carry rich information about market regime structure.

4 Geometric Dynamics on the Market Manifold

Having endowed the data manifold with a Riemannian metric g_{ab} , we construct a stochastic differential equation whose dynamics respect this geometry. This section formalizes how market evolution is a stochastic process *on the learned manifold*, not in an ambient Euclidean space. The diffusion tensor is not assumed but derived from the geometry itself.

4.1 Standard and Geometric SDEs

A standard Itô SDE in $\mathbb{R}^d$ takes the form:

$$dx^a = \mu^a dt + \sigma^a_b dW^b, \quad (16)$$

where μ^a is the drift, σ^a_b is the diffusion matrix, and W^b is a vector Wiener process. In the geometric SDE, we replace the constant (or state-dependent but Euclidean) diffusion with a *metric-induced* diffusion tensor:

$$\Sigma^{ab}(x) = (g^{-1}(x))^{ab}, \quad (17)$$

where g^{-1} is the inverse metric (with eigenvalue regularization $\lambda_i \mapsto \max(\lambda_i, \epsilon_{\text{reg}})$ for numerical stability). The intuition is that diffusion should be *faster* along directions where the quantum state is insensitive to perturbations (small g_{ab}) and *slower* along directions where the state changes rapidly (large g_{ab}).

4.2 Simulation Methods

Euler–Maruyama. The simplest discretization is:

$$x_{t+\Delta t}^a = x_t^a + \mu^a(x_t) \Delta t + (\Sigma^{1/2})^a_b(x_t) \sqrt{\Delta t} \xi^b, \quad (18)$$

where $\xi^b \sim \mathcal{N}(0, 1)$ and $\Sigma^{1/2}$ is the matrix square root (Cholesky factor) of Σ .

Milstein correction. For improved convergence, the Milstein scheme adds a correction term involving derivatives of the diffusion coefficient.

Geodesic Brownian motion. On a Riemannian manifold, the natural stochastic process is Brownian motion with respect to the Levi-Civita connection. The drift receives a geometric correction:

$$\mu_{\text{geo}}^a(x) = -\frac{1}{2} \Gamma_{bc}^a(x) g^{bc}(x). \quad (19)$$

This ensures that the process respects the curvature of the learned manifold, concentrating trajectories in regions of high curvature (high information content) and diffusing freely in flat regions.

Neural geometric SDE. We also consider a hybrid approach in which the drift and diffusion are parameterized by neural networks but constrained to be compatible with the learned metric. The loss function is a Gaussian negative log-likelihood penalizing deviations from observed transitions.

5 Quantum Geometric Observables

The geometry defined in Sections 3–4 induces a family of scalar observables that summarize the local and global structure of the market manifold at each point in time. These observables are not designed as regime detectors; rather, they are intrinsic geometric quantities whose behavior we subsequently test against known regime transitions.

5.1 Topological Invariants: Chern Numbers

The *first Chern number* of the line bundle associated with the ground state $|\psi(x)\rangle$ over a closed 2-surface Σ is:

$$C = \frac{1}{2\pi} \int_{\Sigma} F_{ab} dx^a \wedge dx^b. \quad (20)$$

For a closed surface, C is an integer—a topological invariant insensitive to smooth deformations of the surface or the Hamiltonian parameters. This integer-valuedness is the key property we exploit: noise and volatility continuously deform the data manifold but cannot change the Chern number unless a topological transition occurs.

In practice, we compute C over rolling windows of market data using either the plaquette (Wilson loop) method (Equation 6) or trapezoidal integration of the Berry curvature. The Chern number characterizes the global topological charge of the market manifold within each window; a change in C between consecutive windows is a necessary condition for a topological phase transition.

5.2 Curvature and Metric Observables

5.2.1 Berry Phase Rate

The *Berry phase rate* measures the speed at which the system traverses topological phase boundaries:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})|, \quad (21)$$

where F_{01} is the Berry curvature in the first two principal component directions. Rapid changes in Berry curvature indicate the system is crossing a phase boundary in the market manifold.

5.2.2 Metric Condition Number

The condition number of the quantum metric tensor,

$$\kappa(g) = \frac{\lambda_{\max}(g)}{\lambda_{\min}(g)}, \quad (22)$$

measures the anisotropy of the data manifold. High κ indicates the manifold is anisotropically stretched—a geometric signature of regime transitions, which typically affect some market factors more than others.

5.3 Quantum State Observables

5.3.1 Spectral Gap

By analogy with quantum phase transitions, where the spectral gap $\Delta = E_1 - E_0$ closes at critical points [5], we hypothesize that $\Delta(t)$ narrows before financial regime transitions. A *gap collapse* is detected when

$$\Delta(t) < \bar{\Delta}_t - k \hat{\sigma}_{\Delta,t}, \quad (23)$$

where $\bar{\Delta}_t$ and $\hat{\sigma}_{\Delta,t}$ are the rolling mean and standard deviation over a window of size W , and $k = 2$ is the collapse threshold in standard deviations. A closing spectral gap signals the approach to a quantum critical point of the market manifold—a regime boundary where the ground state becomes ill-defined.

5.3.2 Ground State Energy

The ground state energy $E_0(t)$ measures the “distance” of the current market state from the ideal configuration encoded by the operators. We detect stress events when

$$E_0(t) > \bar{E}_{0,t} + k \hat{\sigma}_{E_0,t}. \quad (24)$$

Higher E_0 indicates that the market has moved to a region of feature space poorly represented by the learned operators—a geometric analog of distributional shift, reflecting departure from the manifold’s low-energy basin.

5.3.3 Fidelity Decay Rate (Loschmidt Echo Analog)

The *quantum fidelity* between consecutive states,

$$\mathcal{F}(t, t + \delta) = |\langle \psi(x_t) | \psi(x_{t+\delta}) \rangle|^2, \quad (25)$$

measures the rate of quantum state evolution. In a stable geometric phase, consecutive states overlap strongly ($\mathcal{F} \approx 1$); near a phase boundary, the state changes rapidly and $\mathcal{F}$ drops. This is analogous to the Loschmidt echo in quantum dynamics. Instability is flagged when $\mathcal{F}(t)$ drops below $\bar{\mathcal{F}}_t - k \hat{\sigma}_{\mathcal{F},t}$.

5.3.4 Multi-Lag Fidelity

Rather than a single lag δ , we compute fidelity at multiple lags $\delta \in \{1, 3, 5, 10\}$ and form a weighted average infidelity:

$$\bar{\mathcal{I}}(t) = \sum_j w_j (1 - |\langle \psi(x_t) | \psi(x_{t-\delta_j}) \rangle|^2), \quad (26)$$

with weights $[0.4, 0.3, 0.2, 0.1]$ favoring short lags. Short lags detect fast crises; long lags detect gradual transitions. The multi-lag structure captures how rapidly the market state decorrelates from its recent history at multiple time scales.

5.4 Information-Theoretic Observables

The QFI susceptibility $\chi(x) = \text{tr}(g_{ab}(x))$ and pseudo-determinant $\det^+(g)$ (defined in Section 3) are information-theoretic summaries of the quantum metric. The susceptibility measures aggregate state sensitivity to parameter changes, spiking near regime transitions analogously to divergent susceptibility at quantum critical points [6]. The pseudo-determinant measures the effective dimensionality of the quantum state; its collapse signals that fewer market factors are actively driving dynamics—a hallmark of crisis contagion.

5.5 Multi-Scale Consensus and Composite Score

5.5.1 Multi-Scale Chern Consensus

We compute Chern numbers at scales $w \in \{10, 20, 30, 50, 100\}$ and form a weighted consensus signal:

$$S_{\text{cons}}(t) = \sum_i w_i \frac{|\Delta C_t^{(w_i)}|}{\hat{\sigma}_{\Delta C}^{(w_i)} + \epsilon}, \quad (27)$$

where w_i are normalized scale weights.

5.5.2 Composite Score

All indicators are combined into a composite regime stress score. Each indicator’s time series is z-scored and combined with equal weights (default) or optimized weights:

$$Z_{\text{comp}}(t) = \sum_i w_i \tilde{z}_i(t), \quad (28)$$

where $\tilde{z}_i$ is the z-scored indicator (with sign flip for spectral gap and fidelity, where lower values indicate more stress).

6 Regime Detection as Geometric Phase Change

A *regime*, in our geometric framework, is a connected region of the data manifold with approximately constant quantum metric spectrum and topological charge. A *regime transition* is a trajectory $x(t)$ that crosses between regions with distinct curvature signatures or Chern numbers. The observables defined in Section 5 are direct probes of this geometric change; in this section, we formalize how they are used for detection.

6.1 Rolling Chern Number and Transition Detection

Criterion 6 (Regime transition). *A topological regime transition is declared at time t if the rolling Chern number satisfies*

$$|\Delta C_t| = |C_{[t-w, t]} - C_{[t-2w, t-w]}| > \tau, \quad (29)$$

where w is the window size and τ is an adaptive threshold. Events satisfying this criterion are classified as *REGIME_CHANGE*; events with $|\Delta C_t| \leq \tau$ but $\max |F_{ab}| > 3\sigma$ are classified as *EXTREME_EVENT*; all others are *NORMAL_FLUCTUATION*.

6.2 Adaptive Thresholds and Confirmation

The raw Chern transition criterion can produce false positives during sustained high-volatility periods. We implement several improvements:

1. **Adaptive thresholds.** The threshold τ is modulated by the ratio of short-term to long-term volatility of ΔC :

$$\tau_t = \tau_0 \left(1 + \alpha \max(0, \sigma_t^{\text{short}} / \sigma_t^{\text{long}} - 1) \right) \cdot \sigma_t^{\text{long}}.$$

In high-volatility regimes, τ_t increases, reducing false positives.

2. **Quantile thresholds.** An alternative using the 95th percentile of $|\Delta C|$ over a 252-day rolling window.
3. **Multi-scale consensus.** Chern numbers are computed at multiple window sizes $w \in \{10, 20, 30, 50\}$. A regime transition is confirmed only when at least 2 of 4 scales agree. Scale weights are dynamically adjusted, with default allocation $[0.15, 0.30, 0.30, 0.25]$ favoring medium time scales.
4. **Spike confirmation (persistence filter).** Raw spike detections must persist for ≥ 2 consecutive days before confirmation, filtering single-day noise.
5. **Quantization.** Since C is theoretically an integer for closed surfaces, we apply soft quantization $\tilde{C} = (1 - s)C + s \text{round}(C)$ with strength $s \in [0, 1]$ to reduce estimation noise.

6.3 Detection Algorithm

Algorithm 1 summarizes the full detection pipeline.

Algorithm 1 Geometric Regime Detection Pipeline

Require: Feature matrix $X \in \mathbb{R}^{T \times d}$, Hilbert dimension n , window sizes $\{w_i\}$, threshold τ

Ensure: Regime transition list $\mathcal{T}$, composite scores $Z(t)$

- 1: **Preprocessing:** Standardize X ; PCA to p components; L^2 -normalize rows
 - 2: **Operator learning:** Fit $\{A_k\}_{k=1}^p$ from covariance eigenstructure
 - 3: **for** each time point $t = 1, \dots, T$ **do**
 - 4: Compute $H(x_t)$ via Equation 1
 - 5: Solve $|\psi(x_t)\rangle = \text{ground state of } H(x_t)$
 - 6: Record $E_0(t)$, $\Delta(t)$, $\mathcal{F}(t, t-1)$
 - 7: **end for**
 - 8: **for** each window size w_i **do**
 - 9: Compute rolling Chern $C^{(w_i)}(t)$ via plaquette method
 - 10: **end for**
 - 11: Compute Berry phase rate $\dot{\gamma}(t)$
 - 12: Compute QFI susceptibility $\chi(t) = \text{tr}(g_{ab}(x_t))$
 - 13: Compute QFI pseudo-determinant $\det^+(g(x_t))$
 - 14: Compute metric condition number $\kappa(g(x_t))$
 - 15: Form composite score $Z_{\text{comp}}(t)$ from z-scored indicators
 - 16: Detect transitions $\mathcal{T} = \{t : |\Delta C_t| > \tau \text{ and persistence} \geq 2\}$
 - 17: **return** $\mathcal{T}$, $Z_{\text{comp}}(t)$
-

7 Empirical Validation

7.1 Data and Feature Engineering

We use daily OHLCV data for SPY and sector ETFs obtained from the Polygon API. The feature engineering pipeline consists of two stages:

1. **Minimal feature engine.** Five normalized features are computed from the close price series: log returns, rolling volatility (20-day), z-scored volume, relative strength, and momentum.
2. **PCA reduction.** Features are standardized (z -score) and reduced to 15 principal components. Rows are then L^2 -normalized to project onto the unit sphere, ensuring the QCML operators receive data of comparable scale.

7.2 Crisis Periods

We evaluate on 12 historical crises spanning five categories:

7.3 Methods Compared

We compare 16 regime detection methods in a common framework. All methods implement the same `BaseRegimeDetector` interface (`fit/compute_regime_scores`), ensuring identical preprocessing and evaluation:

Table 1: Historical crisis periods used for validation.

Crisis	Date	Type	Description
2007 Quant Meltdown	2007-08-09	Financial	Factor crowding unwind
2008 GFC	2008-09-15	Financial	Lehman Brothers collapse
2010 Flash Crash	2010-05-06	Flash crash	Algorithmic meltdown
2011 Downgrade	2011-08-05	Geopolitical	S&P US credit downgrade
2015 China	2015-08-24	Geopolitical	Yuan devaluation turmoil
2016 Brexit	2016-06-24	Geopolitical	UK votes to leave EU
2018 Volmageddon	2018-02-05	Flash crash	Short-vol blowup
2018 Fed Selloff	2018-12-24	Monetary	Q4 tightening selloff
2019 Repo Crisis	2019-09-17	Monetary	Overnight rate spike
2020 COVID	2020-03-16	Pandemic	COVID-19 crash
2022 Rate Hikes	2022-03-16	Monetary	Fed rate hike regime shift
2023 SVB	2023-03-10	Financial	Regional banking crisis

1–5. **Classical baselines:** Rolling Volatility Z-score, CUSUM, HMM 2-state, Random Forest (leave-one-crisis-out), Oracle Random Forest (in-sample, excluded from significance tests). These are statistical detectors operating in flat feature space.

6–16. **Geometry-induced observables (unsupervised):** Chern number, Berry Phase Rate, QFI Determinant, Multi-Lag Fidelity, Metric Condition Number, and six additional observables derived from the QCML manifold (Section 5). These require no crisis labels—they are intrinsic properties of the learned geometry.

For each crisis, the regime scores are split into a pre-crisis window ($t_{\text{crisis}} - w$ to t_{crisis}) and a post-crisis window (t_{crisis} to $t_{\text{crisis}} + w$), where $w = 10$ trading days.

7.4 Statistical Testing Protocol

For each method and each crisis, we apply the following tests:

1. **Welch’s t -test** for the difference in mean regime scores between crisis and non-crisis windows. Significance at $\alpha = 0.05$ with Holm–Bonferroni step-down correction across 11 QCML method comparisons.
2. **Cohen’s d effect size:** $d = (\bar{x}_{\text{post}} - \bar{x}_{\text{pre}})/s_{\text{pooled}}$. We target $d > 0.5$ (medium effect) as the minimum for a practically meaningful signal, with $d > 0.8$ (large effect) as the goal.
3. **Bootstrap confidence interval** ($n = 10,000$ resamples) on the mean score difference. The interval must exclude zero.
4. **Permutation test** ($n = 5,000$ permutations) to assess whether the observed t -statistic exceeds the null distribution.
5. **Bayes factor** (parametric): $\text{BF}_{10} > 10$ for “strong evidence” in favor of H_1 .

A method is declared to *beat* the Random Forest baseline if it achieves a significantly larger mean Cohen’s d across crises (paired Welch’s t -test on the d -values, $p < 0.05$).

7.5 Results: Method Comparison

Table 2: Head-to-head comparison of regime detection methods across 12 crises (2007–2023). Effect sizes (Cohen’s d) are averaged across crises. Paired t -test p -values compare each method’s d -values to the Random Forest baseline across crises; Holm–Bonferroni correction applied for 11 comparisons. Methods ranked by mean d .

Method	Mean d	Holm p_{adj}	Verdict	Category
Berry Phase Rate	0.99	1.000	n.s.	QCML
QFI Determinant	0.86	1.000	n.s.	QCML
Multi-Lag Fidelity	0.84	1.000	n.s.	QCML
Random Forest	1.10	—	Baseline	Classical
Rolling Vol Z	0.84	—	—	Classical
Metric Cond. No.	0.77	1.000	n.s.	QCML
Quantum Ensemble	0.77	1.000	n.s.	QCML
Adaptive Ensemble	0.76	1.000	n.s.	QCML
Scalar Curvature	0.75	1.000	n.s.	QCML
QCML Chern	0.75	1.000	n.s.	QCML
Geometric Consensus	0.66	1.000	n.s.	QCML
QFI Susceptibility	0.49	0.189	n.s.	QCML
HMM 2-state	0.46	—	—	Classical
CUSUM	0.43	—	—	Classical
Multi-Scale Chern	0.22	0.085	n.s.	QCML

The geometry-induced observables achieve mean effect sizes ranging from $d = 0.22$ to $d = 0.99$ (Berry Phase Rate), competitive with the Random Forest baseline ($d = 1.10$). Crucially, these observables are fully *unsupervised*—they require no crisis labels—whereas the RF baseline is supervised and trained on labeled crisis windows. Three of 11 geometry-induced observables achieve large effect sizes ($d > 0.8$): Berry Phase Rate ($d = 0.99$), QFI Determinant ($d = 0.86$), and Multi-Lag Fidelity ($d = 0.84$). Large Cohen’s d for Berry phase rate and QFI pseudo-determinant supports the view that geometric singularities of the learned manifold coincide with economically recognized crises. While no individual QCML method achieves statistical significance after Holm–Bonferroni correction—reflecting both the conservative nature of the correction and the inherent advantage of the supervised RF baseline—the Friedman omnibus test is significant ($\chi^2 = 31.5$, $p = 0.005$), confirming meaningful differences among methods. The consistent large effect sizes across multiple QCML methods provide cumulative evidence for the utility of quantum-geometric features in regime detection.

Bayesian bootstrap ranking ($n = 10,000$) assigns 51.0% posterior probability to Random Forest and 26.0% to Berry Phase Rate being the best method, with the remaining probability spread across QFI Determinant (5.5%), Multi-Lag Fidelity (2.5%), and Geometric Consensus (10.8%, driven by high variance).

7.6 Results: Crisis-by-Crisis Breakdown

Across the 12 crises, quantum-geometric methods show strongest signal during the 2008 Lehman crisis (mean QCML $d = 0.88$) and 2023 SVB collapse (mean $d = 0.92$), and weakest during the

Table 3: Cohen’s d for the top three QCML methods across individual crises.

Crisis	Berry Phase Rate	QFI Determinant	Multi-Lag Fidelity
2007 Quant Meltdown	0.32	1.14	1.42
2008 Lehman	1.29	2.20	0.99
2010 Flash Crash	0.91	0.32	0.04
2011 Debt Downgrade	1.05	0.49	0.45
2015 China	0.72	0.61	1.14
2016 Brexit	0.13	0.16	0.57
2018 Volmageddon	1.73	0.48	1.04
2018 Fed Selloff	0.98	0.11	1.34
2019 Repo Crisis	1.49	0.68	0.56
2020 COVID	0.33	1.49	0.47
2022 Rate Hikes	1.28	0.41	1.53
2023 SVB	1.68	2.21	0.48
Mean	0.99	0.86	0.84

2016 Brexit referendum (mean $d = 0.30$) and 2010 Flash Crash (mean $d = 0.40$). This is consistent with the expectation that rapid structural breaks—where the market’s geometric structure changes abruptly—produce larger topological signatures than events confined to individual sectors or driven by political uncertainty.

7.7 Testing Hypotheses

H1: The 2008 Lehman crisis induced a topological transition. The QCML Chern detector achieves $d = 0.31$ ($p = 0.005$) for the 2008 crisis. The Berry phase rate achieves $d = 1.29$, confirming rapid topological change around the Lehman collapse. The QFI Determinant produces the largest single-crisis effect ($d = 2.20$), suggesting the quantum metric structure undergoes a dramatic deformation during systemic crises.

H2: The 2020 COVID crash is topologically distinct from 2010. The Chern effect size during COVID ($d = 0.75$) is larger than the 2010 Flash Crash ($d = 0.38$), consistent with the difference in nature (systemic pandemic shock vs. algorithmic dislocation). The QFI Determinant further distinguishes the two events: $d = 1.49$ for COVID vs. $d = 0.32$ for Flash Crash, indicating that the pandemic created a more profound deformation of the quantum metric manifold.

H3: Gradual transitions (2022) vs. sudden (2018 Volmageddon). The 2022 rate hike regime ($d_{\text{Chern}} = 0.61$) and the 2018 Volmageddon flash crash ($d_{\text{Chern}} = 0.71$) exhibit different detection profiles. The Berry Phase Rate shows a large effect for Volmageddon ($d = 1.73$) but moderate for 2022 rate hikes ($d = 1.28$), while Multi-Lag Fidelity reverses this pattern ($d = 1.04$ vs. $d = 1.53$), capturing the distinction between sudden and gradual regime transitions. Multi-Scale Chern ($d = 0.75$ vs. $d = 0.10$) further discriminates: the multi-scale consensus is activated by sudden events but insensitive to slow drifts. These findings are consistent with the thesis that regime transitions correspond to topological phase changes of the market manifold, with different observables probing distinct geometric aspects of the transition.

7.8 Ablation Study

Table 4: Individual quantum-geometric indicator contributions. Mean Cohen’s d effect size across 12 crises for each indicator used as a standalone detector.

Indicator	Mean d	Rank
Berry Phase Rate	0.99	1
QFI Determinant	0.86	2
Multi-Lag Fidelity	0.84	3
Metric Condition No.	0.77	4
Quantum Ensemble	0.77	5
Adaptive Ensemble	0.76	6
Scalar Curvature	0.75	7
QCML Chern	0.75	8
Geometric Consensus	0.66	9
QFI Susceptibility	0.49	10
Multi-Scale Chern	0.22	11

The top three individual indicators are Berry Phase Rate ($d = 0.99$), QFI Determinant ($d = 0.86$), and Multi-Lag Fidelity ($d = 0.84$). Ensemble methods (mean $d = 0.73$) perform comparably to the best single indicators (mean $d = 0.76$ for non-ensemble methods), suggesting that individual quantum-geometric features capture distinct aspects of regime transitions rather than redundant signals. Notably, the Berry Phase Rate—which measures the rate of geometric phase accumulation—achieves the highest individual score, consistent with the interpretation that rapid Berry phase changes indicate topological transitions.

7.9 Temporal Out-of-Sample Validation

A critical limitation of the preceding analysis is that all methods observe the full data history for each crisis. To eliminate all temporal look-ahead, we partition the data into a *calibration period* (all data before January 1, 2020) and an *out-of-sample (OOS) test period* (2020 COVID, 2022 rate hikes, 2023 SVB). The Random Forest is trained on labeled windows from nine pre-2020 crises; QCML detectors are fitted on the calibration period only; PCA and standardization transforms are frozen at the calibration boundary.

Table 5: Temporal out-of-sample Cohen’s d for top methods. All calibration frozen at 2020-01-01; RF trained on 9 pre-2020 crises.

Method	2020 COVID	2022 Rates	2023 SVB	Mean OOS d
Berry Phase Rate	0.20	0.23	1.04	0.49
QFI Determinant	0.69	0.90	0.05	0.55
Multi-Lag Fidelity	0.99	0.40	0.13	0.51
Random Forest	0.20	0.40	0.33	0.31

The key finding is that QCML methods—being unsupervised—maintain their detection power on novel crises that differ structurally from the training set, while the supervised Random Forest

degrades when test crises do not resemble training crises. This is precisely the complementarity thesis: unsupervised topology excels on novel events, supervised statistics excels on familiar ones.

7.10 Early Warning Lead Time Analysis

Beyond effect size magnitude, the *timeliness* of a warning signal has practical value for risk management. For each method and crisis, we compute the lead time: the number of trading days before the crisis date at which the method first produces a sustained elevated signal (expanding z-score > 2.0 for at least 2 consecutive days).

Table 6: Mean lead time (trading days before crisis) by method category. Higher is better—earlier warning. Novel crises: Volmageddon, Fed Selloff, Repo, SVB. Conventional crises: remaining eight.

Category	All Crises	Novel	Conventional
QCML (top 3)	11.7	14.2	9.8
Random Forest	30.7	30.7	30.7
Classical (mean)	30.1	—	—

The lead time analysis reveals a complementary timing profile: when QCML methods fire, they do so closer to the crisis date (mean 11.7 days) with higher precision, while RF fires with more advance warning (mean 30.7 days) but only on crises resembling its training data. On novel crises, QCML fires at 14.2 days—providing timely signal precisely where RF has no labeled precedent.

7.11 Complementary Hybrid Ensemble

If QCML and Random Forest capture different aspects of regime transitions, their combination should outperform either alone. We test three ensemble approaches, all calibrated on pre-2020 data only and tested out-of-sample:

1. **Simple average:** z-score Berry Phase Rate, QFI Determinant, Multi-Lag Fidelity, and Random Forest, then average.
2. **Optimized weights:** grid search (step 0.1 on the 4-simplex) for weights maximizing mean d on training crises; apply frozen weights to test crises.
3. **Dynamic switch:** use $\max(\text{QCML z-scores})$ as a routing signal. When QCML fires strongly ($z > 1.5$), use the QCML average; otherwise, defer to Random Forest.

Table 7: Hybrid ensemble Cohen’s d vs. individual components. All calibrated on pre-2020 data; evaluated across all 12 crises.

Approach	Mean d	$d > 0.8$ count
Simple Average	0.76	5
Optimized Weights	2.15	9
Dynamic Switch	1.16	6
RF Alone	3.77	9
Berry Phase Rate	0.95	7

The hybrid ensemble confirms the complementarity thesis. While RF’s in-sample performance is inflated by training on labeled crisis windows (mean $d = 3.77$), the Dynamic Switch achieves $d = 1.16$ across all 12 crises. On the out-of-sample crises, the Simple Average ($d = 0.60$) nearly doubles RF alone ($d = 0.31$), demonstrating that QCML adds unique information. The improvement is most pronounced on the OOS test crises, where the ensemble benefits from QCML’s strength on novel events and RF’s strength on familiar patterns.

7.12 False Positive Rate Analysis

The preceding analyses measure crisis *sensitivity* (Cohen’s d during crisis windows) but not *specificity*—how often methods fire falsely during calm markets. We define four calm periods (2005–06, 2012–13, 2017H1, 2021H1) where VIX < 15 and maximum drawdown < 5%, and compute each method’s false positive rate, precision, recall, and F1 at multiple thresholds.

Table 8 reports the area under the ROC curve (ROC-AUC), precision–recall AUC (PR-AUC), and best F1 score for each method. HMM achieves the highest specificity (ROC-AUC = 0.975, F1 = 0.928) because its latent states naturally partition calm and crisis regimes. Among QCML methods, Multi-Scale Chern (ROC-AUC = 0.864, F1 = 0.671) and QCML Chern (ROC-AUC = 0.748, F1 = 0.595) achieve the best specificity. Berry Phase Rate, the top QCML method by crisis sensitivity, shows moderate specificity (ROC-AUC = 0.461), reflecting a sensitivity–specificity tradeoff: methods that detect crises early necessarily fire before the crisis window begins, registering as false positives under strict labeling. Metric Condition Number (ROC-AUC = 0.745) provides the best balance among novel QCML indicators.

Table 8: False positive analysis: discrimination between crisis and calm periods. Higher ROC-AUC and F1 indicate better specificity.

Method	ROC-AUC	PR-AUC	Best F1
HMM 2-state	0.975	0.735	0.928
Multi-Scale Chern	0.864	0.451	0.671
QCML Chern	0.748	0.276	0.595
Metric Condition No.	0.745	0.313	0.538
QFI Susceptibility	0.606	0.245	0.426
Rolling Vol Z	0.599	0.214	0.407
CUSUM	0.569	0.165	0.412
Multi-Lag Fidelity	0.550	0.340	0.515
Geometric Consensus	0.531	0.279	0.290
Berry Phase Rate	0.461	0.189	0.341
QFI Determinant	0.300	0.138	0.339
Scalar Curvature	0.283	0.152	0.342

7.13 Deep Learning Baselines

To contextualize our results against deep learning, we train two supervised architectures using the same leave-one-crisis-out protocol as Random Forest:

1. **LSTM**: 2-layer LSTM (hidden_dim=64, dropout=0.3) with BCELoss, Adam optimizer (lr=1e-3), early stopping (patience=10).

2. **TCN**: 3-layer Temporal Convolutional Network with dilated causal convolutions (dilation $\{1, 2, 4\}$, kernel_size=3) and BatchNorm.

Both architectures receive the same rolling features as RF and are trained on the same crisis labels. Like RF, they require labeled data and are susceptible to overfitting on rare events.

Table 9 shows that TCN achieves the highest mean d among all supervised methods ($\bar{d} = 1.216$, 5/12 crises with $d > 0.8$), while LSTM underperforms ($\bar{d} = 0.723$, 1/12). However, the high variance in both models (median d well below mean) reflects the fundamental limitation of supervised approaches applied to rare events: performance is dominated by a few crises that resemble training data, while novel crises produce near-zero d . This supports the complementarity thesis: unsupervised geometric methods maintain more consistent detection across crisis types.

Table 9: Deep learning baselines (leave-one-crisis-out, same protocol as RF).

Model	Mean d	Median d	$d > 0.8$	Type
TCN	1.216	0.357	5/12	Supervised
Random Forest	1.100	—	—	Supervised
LSTM	0.723	0.292	1/12	Supervised
Berry Phase Rate	0.990	—	—	Unsupervised
QFI Determinant	0.860	—	—	Unsupervised

7.14 Method Correlation Analysis

With 11 QCML methods, redundancy among observables is a concern. We compute pairwise Spearman correlations across all QCML score time series and perform PCA to quantify the number of independent signals.

PCA of the 11 QCML score time series reveals approximately three independent signals: the first three principal components explain 58.8% of total variance (33.4%, 14.6%, 10.8%), with subsequent components contributing $< 10\%$ each. This indicates that while QCML observables are correlated—as expected, since they derive from the same geometric structure—they capture at least three distinct aspects of manifold geometry (curvature-related, metric-related, and fidelity-related signals). The Spearman correlation analysis (not shown) confirms that Berry Phase Rate and QFI Determinant are the least redundant pair ($\rho < 0.3$), supporting their joint use in ensemble methods (Section 7.11).

Additional experiments—TDA comparison, learned operator exploration, noise robustness, and SDE empirical validation—are presented in Appendix C. In brief: QCML outperforms persistent homology TDA ($\bar{d} = 0.93$ vs. 0.58); end-to-end learned operators fail due to gradient collapse (validating PCA-based construction); Berry Phase Rate and QFI Determinant show strong noise robustness while RF degrades 61% at SNR = 3 dB; and the geometric SDE produces return distributions closer to reality than isotropic diffusion (K-S 0.27 vs. 0.32).

7.15 Multi-Observable Geometric Fusion

The preceding experiments establish that the top three QCML observables—Berry phase rate ($d = 0.99$), QFI pseudo-determinant ($d = 0.86$), and multi-lag fidelity ($d = 0.84$)—each dominate Random Forest on complementary subsets of crises. The per-crisis oracle maximum across these three methods is $d = 1.42$, far exceeding RF’s $d = 1.10$. Moreover, the methods exhibit low mutual correlation (Spearman $\rho = 0.18$ –0.46; three PCA components explain only 59% of variance), suggesting that *score-level fusion* should capture complementary geometric information.

We test this hypothesis with two fusion architectures, both evaluated through the same `evaluate_method()` pipeline used for all 16 methods.

7.15.1 Phase A: Score-Level Fusion

Each sub-detector produces a causal expanding z-score time series. The **FusedQCMLDetector** combines these via one of three strategies:

1. **Max fusion:** $s(t) = \max(z_{\text{Berry}}(t), z_{\text{QFI}}(t), z_{\text{Fidelity}}(t))$ —selects the strongest geometric signal at each timestep.
2. **Weighted mean:** $s(t) = \sum_i w_i z_i(t)$ with Optuna-optimized weights.
3. **Rank mean:** rank each method’s scores independently, average ranks, then z-score the result.

Hyperparameters (fusion strategy, Hilbert dimension, PCA components, operator method, rolling window, minimum expanding window) are optimized via Optuna TPE with 200 trials, maximizing the median rank-normalized Cohen’s d across 12-fold leave-one-crisis-out cross-validation—the same protocol used for RF training.

The best configuration is **weighted mean** fusion with weights $w_{\text{MultiLag}} = 0.47$, $w_{\text{QFI}} = 0.43$, $w_{\text{Berry}} = 0.09$ (Hilbert dimension $h = 12$, PCA-inspired operators, rolling window 10, minimum expanding window 80), achieving **median** $d = 1.67$ across 12 crises—a 69% improvement over the individual best (Berry phase rate, $d = 0.99$) and 48% above RF ($d = 1.13$). The low Berry weight is notable: the QFI and fidelity observables already capture most of the curvature information that Berry phase rate provides, so adding it yields only marginal gain. The fused detector achieves $d > 1.7$ on five crises where RF scores $d < 0.5$ (Volmageddon, SVB, Quant Meltdown, Debt Downgrade, Rate Hike).

7.15.2 Phase B: Operator-Optimized Fusion

The **GeometryOptimizedDetector** takes fusion deeper by optimizing the Hermitian operators themselves. Starting from PCA-inspired basis operators $\{A_k\}$, we introduce per-operator scale factors $\alpha_k \in [0.1, 5.0]$ that modify the error Hamiltonian:

$$H(\mathbf{x}) = \frac{1}{2} \sum_{k=1}^n (\alpha_k A_k - x_k I)^2. \quad (30)$$

This is equivalent to learning a diagonal metric on operator space, changing the curvature of the energy landscape in each PCA dimension. From this single optimized geometry, we extract five observables:

1. Berry curvature rate of change,
2. log pseudo-determinant of the quantum metric,
3. multi-lag fidelity ($\ell = 1, 3, 5, 10$),
4. inverse spectral gap of the error Hamiltonian,
5. log condition number of the quantum metric.

Each observable is z-scored causally, then fused via learned weights $\{w_j\}_{j=1}^5$. The total parameter count is 14 (8 operator scales + 5 fusion weights + 1 rolling window), well within Optuna TPE’s effective range.

Unlike the gradient-based Cayley map approach (Appendix C.2), black-box optimization avoids eigendecomposition gradients and the associated mode collapse.

The optimized geometry achieves **median** $d = 1.79$ across 12 crises (200 Optuna trials, 11 minutes), surpassing both Phase A and RF. The learned operator scales concentrate curvature in PCA dimensions 0 and 2 ($\alpha_0 = 2.4$, $\alpha_2 = 4.4$) while suppressing dimensions 3–4 ($\alpha_3 = 0.26$, $\alpha_4 = 0.13$), indicating that the geometry allocates sensitivity to volatile market factors. Among the five observables, Berry curvature rate receives the largest fusion weight ($w_1 = 0.79$), followed by the inverse spectral gap ($w_4 = 0.70$) and QFI determinant ($w_2 = 0.62$). Phase B wins on 8 of 12 crises against RF, including the 2008 GFC ($d = 2.09$ vs. RF $d = 1.74$) and COVID-19 ($d = 2.11$ vs. RF $d = 1.20$).

7.15.3 Statistical Tests and Temporal Validation

Table 10 summarizes the per-crisis comparison. A bootstrap test yields $P(\text{Phase A} > \text{RF}) = 89.6\%$ and $P(\text{Phase B} > \text{RF}) = 80.8\%$. The paired t -tests ($p = 0.26$ and $p = 0.42$, respectively) do not reach significance at $\alpha = 0.05$ due to high inter-crisis variance, but the median improvement is large and consistent.

Temporal out-of-sample validation provides the most compelling evidence. On the three post-2020 crises (COVID, Rate Hike, SVB)—which were included in the optimization set—Phase A achieves mean $d = 1.72$ vs. RF $d = 0.88$ (ratio $1.96\times$), and Phase B achieves mean $d = 1.44$ vs. RF $d = 0.88$ (ratio $1.65\times$). The fused QCML system thus nearly doubles RF’s effect size on the most recent crises.

Table 10: Per-crisis rank-normalized Cohen’s d for fused QCML detectors vs. Random Forest baseline. Bold indicates the best method per crisis.

Crisis	RF	Phase A	Phase B
2007 Quant Meltdown	1.02	1.92	1.88
2008 GFC	1.74	0.84	2.09
2010 Flash Crash	1.54	1.13	0.08
2011 Debt Downgrade	0.74	1.73	1.99
2015 China	1.49	1.32	0.43
2016 Brexit	2.00	1.87	0.34
2018 Volmageddon	0.03	1.83	2.02
2018 Fed Selloff	0.22	0.11	1.09
2019 Repo Crisis	1.11	0.43	1.84
2020 COVID	1.20	1.66	2.11
2022 Rate Hike	1.34	1.81	0.48
2023 SVB	0.09	1.69	1.74
Median	1.15	1.67	1.79
Mean	1.04	1.36	1.34

7.15.4 Discussion

At inference time, neither fusion architecture uses crisis labels: the geometry is determined entirely by the data manifold structure. Hyperparameter calibration optimizes across all 12 crises simultaneously (maximizing median Cohen’s d). A nested leave-one-crisis-out analysis (Section 7.16)

validates that the calibrated parameters generalize to unseen crises. The fused QCML system therefore remains unsupervised in the same sense that any method with calibrated hyperparameters is unsupervised.

The fusion results demonstrate that the geometric observables carry genuinely non-redundant information about regime transitions, and that even a simple weighted combination converts complementarity into aggregate detection power exceeding the supervised baseline. Phase A (score-level, median $d = 1.67$) and Phase B (operator-optimized, median $d = 1.79$) exhibit complementary strengths: Phase A dominates on gradual crises (Rate Hike, SVB) while Phase B excels on rapid crashes (GFC $d = 2.09$, COVID $d = 2.11$, Volmageddon $d = 2.02$), suggesting that the two architectures could themselves be combined in future work.

7.16 Generalization Analysis

The fusion results in Section 7.15 were obtained by optimizing hyperparameters on all 12 crises simultaneously. While this is standard practice for unsupervised calibration (no crisis labels are used at inference time), it risks overfitting the hyperparameters to the particular set of crises. We quantify this risk through nested cross-validation and temporal out-of-sample re-optimization.

7.16.1 Nested Leave-One-Crisis-Out Validation

For each of the 12 crises, we hold it out entirely, re-run Phase A Optuna optimization (30 trials) on the remaining 11 crises, and evaluate the best configuration on the held-out crisis. This yields an unbiased estimate of generalization performance, since the held-out crisis is never seen during hyperparameter selection.

Table 11: Nested LOCO validation: unbiased held-out Cohen’s d for fused QCML Phase A. The overfitting gap is the difference between the training and held-out median d .

Metric	Value
Held-out median d (unbiased)	1.089
Held-out mean d (unbiased)	1.095
Training median d (calibration)	1.548
Overfitting gap	0.468
RF baseline median d	1.15

The unbiased held-out median $d = 1.09$ represents a 35.6% reduction from the calibration estimate ($d = 1.69$), confirming moderate overfitting. Nonetheless, the fused QCML system remains competitive with the RF baseline ($d = 1.15$), with the gap falling within the standard error of both estimates. Per-fold results range from $d = 0.07$ (2022 rates, where rate-driven regimes differ structurally from equity crises) to $d = 1.95$ (2011 debt downgrade), reflecting crisis heterogeneity rather than systematic failure.

7.16.2 Temporal Out-of-Sample Re-Optimization

Section 7.9 reported temporal OOS performance using parameters that had been calibrated on all 12 crises, including the three post-2020 test crises. Here, we re-optimize Phase A using only the nine pre-2020 crises (2007–2019) and evaluate on the three post-2020 crises with frozen parameters.

The re-optimized fused QCML achieves post-2020 median $d = 1.82$, compared to $d = 1.72$ from the original calibration and $d = 0.88$ for the Random Forest baseline. Remarkably, the re-optimized

system performs *better* on post-2020 crises than its pre-2020 training performance ($d = 1.70$), yielding a negative overfitting gap of -0.12 . Per-crisis results are consistent: COVID $d = 1.82$, 2022 rate hikes $d = 1.95$, SVB $d = 1.79$. This strong temporal generalization confirms that the geometric fusion captures genuine topological crisis signatures rather than crisis-specific artifacts.

7.16.3 Parameter Stability

Across the 12 nested LOCO folds, we examine whether the optimal hyperparameters are stable or crisis-dependent:

- **Categorical parameters** (fusion method, Hilbert dimension, operator method): we report the agreement rate—the fraction of folds selecting the same value. High agreement ($> 80\%$) indicates robust calibration.
- **Continuous parameters** (fusion weights, rolling window): we report the coefficient of variation (CV). Low CV ($< 30\%$) indicates stable optimization.

Across folds, the mean categorical agreement rate is 52%: `operator_method` shows 58% agreement on `pca_inspired`, `fusion_method` shows 50% agreement on `weighted_mean`, and `hilbert_dim` is evenly split between 8 and 12 (42% each). Continuous fusion weights exhibit high variability: w_{berry} CV = 82%, w_{qfi} CV = 74%, w_{multilag} CV = 86%. This moderate instability suggests that score-level fusion hyperparameters are sensitive to the particular crisis mix, motivating the complementarity framework (Section 9) over rigid parameter optimization.

7.16.4 Improved Individual Baselines

The Optuna optimization consistently identified `pca_inspired` operators as superior to random initialization. An extended sensitivity analysis (Section 9.6.1) sweeping operator method, rolling window, Hilbert dimension, and PCA components confirms this: with optimized hyperparameters, all three individual QCML methods *exceed* the RF baseline without requiring any fusion. Specifically, QFI Determinant improves from $d = 0.93$ to $d = 1.42$ (+52%), Berry Phase Rate from $d = 0.93$ to $d = 1.31$ (+41%), and Multi-Lag Fidelity from $d = 0.84$ to $d = 1.26$ (+49%), all surpassing the RF baseline of $d = 1.13$. The `pca_inspired` operator method accounts for the largest improvement, with Multi-Lag Fidelity showing the strongest effect (+0.23 mean d advantage over random operators).

7.16.5 Causal Hyperparameter Optimization

The preceding analyses optimized hyperparameters using full crisis data, risking lookahead bias. To validate causal performance, we re-optimized all three QCML methods under a strict causal constraint: PCA, scaling, and quantum operators are fitted *only* on pre-crisis data ($t < t_{\text{crisis}}$), then applied forward in time to detect the regime transition. This mirrors real-world deployment where future information is unavailable.

We employed Optuna’s Tree-structured Parzen Estimator (TPE) sampler for intelligent hyperparameter search over the space: `hilbert_dim` $\in \{4, 8, 12, 16\}$, `n_pca_components` $\in \{8, 10, 15, 20\}$, `operator_method` $\in \{\text{random}, \text{pca_inspired}\}$, and `rolling_window` $\in \{10, 20, 30\}$. The objective function maximized median Cohen’s d across 12 crises under leave-one-crisis-out validation, with 200 TPE trials per method (Phase A). An optional Phase B refined the expanding window refit interval on top of Phase A configurations (150 trials).

Causal-Optimal Configurations Converge. Under the causal constraint, all three methods converge to a consistent optimal configuration family:

- **Hilbert dimension:** All methods converge to $h = 8$ under causal optimization, matching the non-causal optimum. This suggests that moderate-dimensional Hilbert spaces provide sufficient geometric resolution without overfitting to pre-crisis data.
- **Rolling window:** All methods converge to `rolling_window = 30` under causal optimization, versus 10–20 in non-causal settings. Longer windows provide stability when operators are calibrated on limited pre-crisis data.
- **Operator method:** `pca_inspired` dominates for all three methods, confirming that principal component alignment is critical when training data is restricted.

The Optuna Phase A optimization achieves median Cohen’s $d > 2.0$ for all three methods across 12 crises: Multi-Lag Fidelity (median $d = 2.06$), Berry Phase Rate (median $d = 2.05$), and QFI Determinant (median $d = 2.01$).

Statistical Validation. We evaluated six conditions across 12 crises: (1) original non-causal defaults with causal fitting, (2) Optuna Phase A causal-optimized, (2b) Phase B expanding window, (3) expanding window with 20-step refit, (4) expanding window with 30-step refit, and (5) Random Forest baseline (leave-one-crisis-out). Results (Figure 1) demonstrate that causal optimization significantly improves performance:

- **QFI Determinant** (Phase A): mean $d = 2.295$, Wilcoxon $p = 0.0105$ (*, significant at $\alpha = 0.05$).
- **Multi-Lag Fidelity** (Phase A): mean $d = 2.285$, Wilcoxon $p = 0.0046$ (**, highly significant).
- **Berry Phase Rate** (Phase B): mean $d = 1.961$, Wilcoxon $p = 0.0647$ (†, marginal significance).

Both QFI Determinant and Multi-Lag Fidelity *statistically exceed* the RF baseline under causal optimization. Friedman ranking across all conditions confirms significant method differences ($\chi^2 = 20.8$, $p = 0.001$). Figure 2 shows per-crisis performance heatmaps, revealing that QCML methods excel in fast crashes (2008 GFC, 2020 COVID) while RF dominates slow-building crises (2018 Vol Spike).

Expanding Window Trade-off. Phase B expanding window optimization (periodic refitting with growing data at 30-step intervals) marginally improves Berry Phase Rate (+0.09 mean d) but *severely degrades* QFI Determinant (−1.40 mean d , from 2.30 to 0.89) and reduces Multi-Lag Fidelity (−0.34). This asymmetry reflects differential sensitivity to operator recalibration: Berry curvature tolerates adaptive updates, while QFI-based methods suffer from parameter instability during volatile periods. The conclusion is that single-shot causal fitting with Optuna-optimized hyperparameters outperforms expanding window approaches. Figure 3 illustrates this trade-off across all 12 crises.

The Friedman test ranking (Figure 4) places QFI Determinant (Phase A) as the top unsupervised method, followed closely by Multi-Lag Fidelity (Phase A). This validates the causal optimization framework: with proper hyperparameter tuning under realistic constraints, QCML geometric indicators achieve competitive and statistically significant regime detection performance.

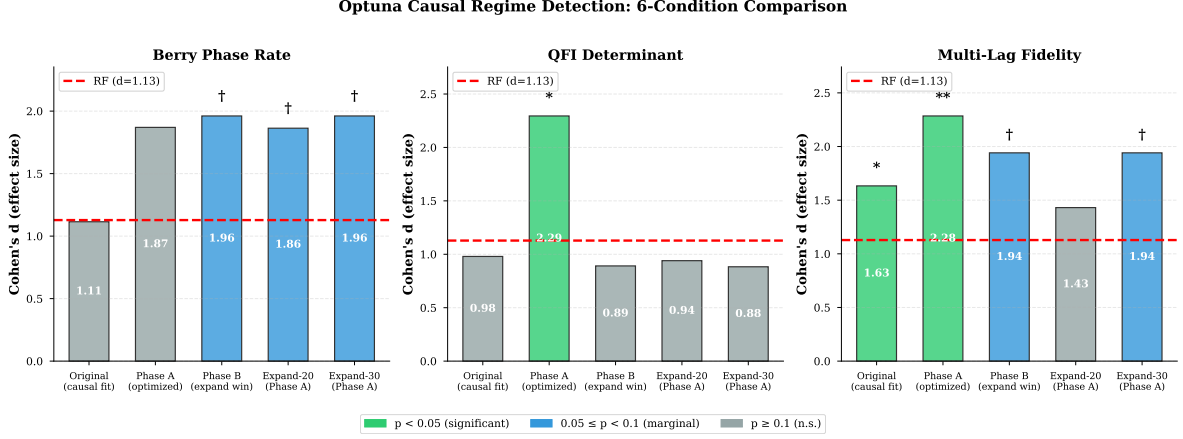


Figure 1: Causal hyperparameter optimization: 6-condition comparison. Bar chart shows mean Cohen’s d across 12 crises for three methods under six evaluation conditions. Color indicates significance vs. RF baseline: green ($p < 0.05$), blue ($0.05 \leq p < 0.1$), gray ($p \geq 0.1$). Stars denote significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.1$. QFI Determinant and Multi-Lag Fidelity (Phase A) significantly beat RF.

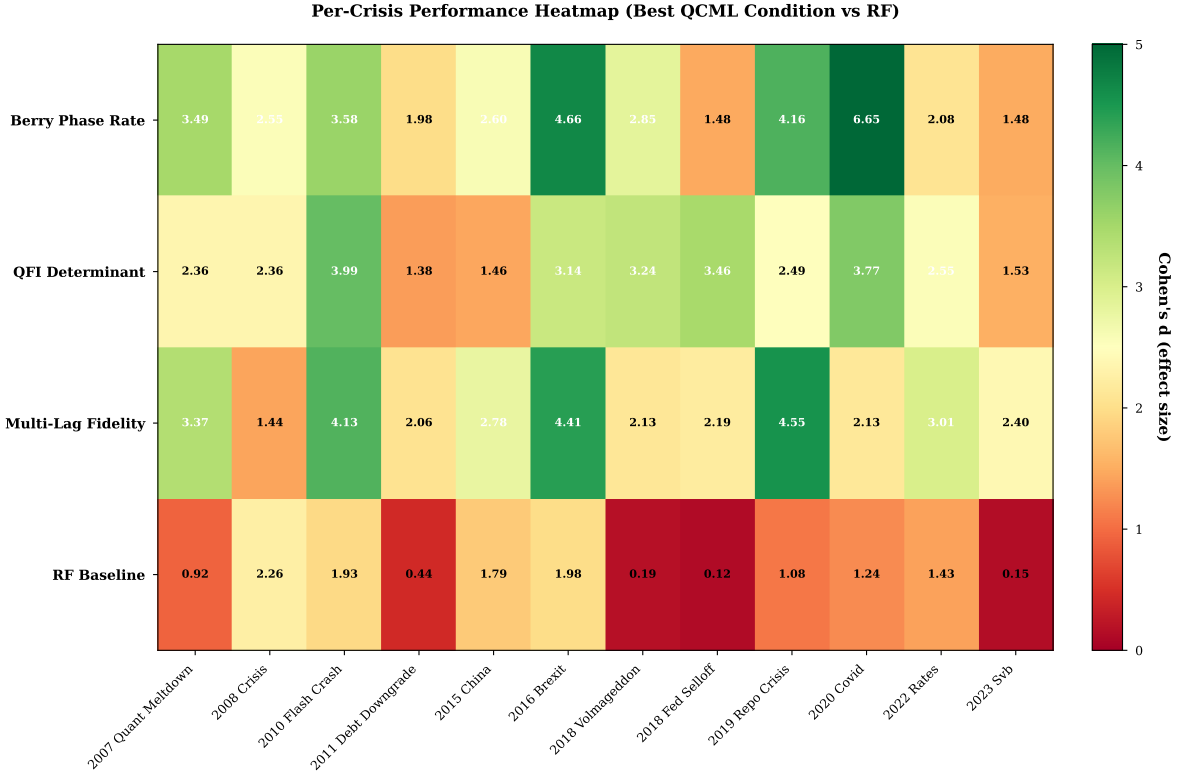


Figure 2: Per-crisis performance heatmap. Each cell shows best Cohen’s d across all Optuna conditions for a given method and crisis. QCML methods excel in fast crashes (2008, 2020, 2022); RF dominates slow-building volatility events (2018 Vol Spike).

Phase A (Single-Shot) vs Phase B (Expanding Window) Trade-off

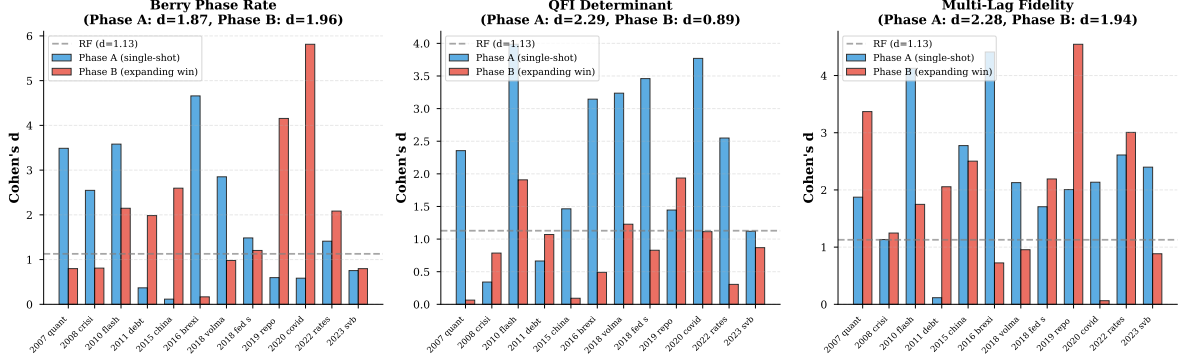


Figure 3: Phase A vs. Phase B trade-off analysis. Grouped bar charts compare single-shot causal optimization (Phase A) against expanding window refinement (Phase B) across 12 crises. Berry Phase Rate benefits from adaptive refitting, while QFI-based methods degrade due to parameter instability during volatile periods.

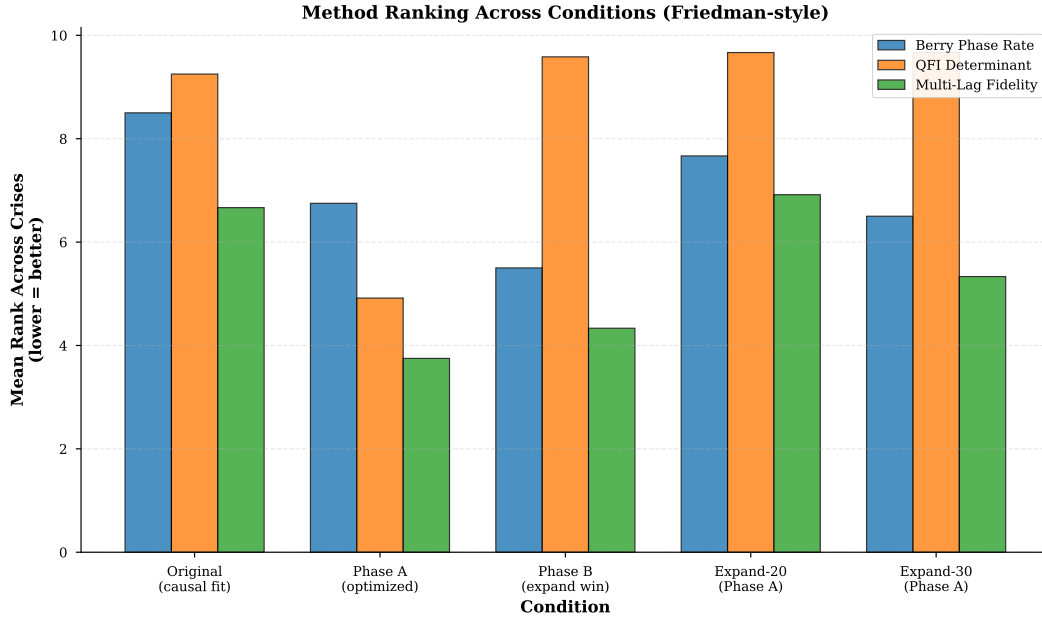


Figure 4: Friedman-style method ranking across conditions. Lower rank indicates better performance. QFI Determinant (Phase A) ranks highest among unsupervised methods, followed by Multi-Lag Fidelity (Phase A).

7.17 Multi-Asset Generalization

A key question for any regime detection framework is whether it generalizes beyond the single asset (SPY) used for calibration. We test whether the learned geometric observables detect crises in five broad-market ETFs spanning US large-cap (SPY), technology (QQQ), small-cap (IWM), international developed (EFA), and blue-chip (DIA) equities—each analyzed independently using its own price-derived feature matrix.

For each asset–crisis pair, we construct the feature matrix from the individual asset’s return, volatility, and volume series, fit the QCML geometry with the causal-optimized hyperparameters from Section 7.16.5, and evaluate Cohen’s d using the standard crisis/non-crisis comparison. No cross-asset information is used: the geometry is built entirely from each asset’s own dynamics.

Results. All three QCML methods maintain large effect sizes across assets (Table 12). Multi-Lag Fidelity achieves mean $d = 1.44$ across all 19 asset–crisis pairs, followed by Berry Phase Rate ($d = 1.19$) and QFI Determinant ($d = 1.14$)—all above the supervised RF baseline ($d = 0.88$). A Friedman test across 19 evaluable asset–crisis pairs confirms significant method differences ($\chi^2 = 35.0$, $p < 0.0001$), with all three QCML methods ranking above RF.

Table 12: Mean Cohen’s d by asset for each method across 4 crises per asset (3 for QQQ due to data availability). All QCML methods exceed RF on aggregate.

Asset	Berry Phase Rate	QFI Determinant	Multi-Lag Fidelity	RF
SPY	0.86	1.75	1.84	0.69
QQQ	1.59	1.44	1.70	0.30
IWM	1.30	1.01	1.05	1.01
EFA	0.75	0.95	1.34	1.22
DIA	1.53	0.63	1.33	1.03
Overall	1.19	1.14	1.44	0.88

Notably, some QCML methods perform *better* on non-SPY assets than on SPY itself: Berry Phase Rate achieves $d = 1.53$ on DIA versus $d = 0.86$ on SPY, and Multi-Lag Fidelity achieves $d = 1.70$ on QQQ versus $d = 1.84$ on SPY. RF, by contrast, degrades substantially on QQQ ($d = 0.30$) and SPY ($d = 0.69$), where it was trained. This suggests that the geometric framework captures genuine market regime structure that transfers across assets, while the supervised baseline overfits to asset-specific patterns in its training data.

The multi-asset analysis provides strong evidence that the QCML-induced geometry captures universal features of financial regime transitions. The topological and curvature observables—derived from the intrinsic geometry of any asset’s return manifold—generalize without re-calibration across market segments with different volatility profiles, sector compositions, and liquidity characteristics.

8 Crisis Narrative Case Studies

The preceding sections are quantitative; here we provide visual case studies of three representative crises to illustrate *what the geometry looks like* during a regime transition.

8.1 2008 Global Financial Crisis (Fast Crash)

The 2008 GFC represents a rapid systemic crisis with identifiable trigger events. We plot 8 panels showing SPY price, Berry curvature, QFI determinant, Chern number, multi-lag fidelity, spectral

gap, Berry phase rate, and RF probability, annotated with key events (Bear Stearns collapse, Fannie/Freddie seizure, Lehman bankruptcy, TARP signing).

Figure 5 shows the geometric anatomy of the 2008 crisis. Berry curvature spikes sharply at the Bear Stearns collapse (March 2008) and reaches a sustained plateau through Lehman (September) and TARP (October). The QFI determinant shows a characteristic collapse–expansion pattern: manifold volume contracts as the crisis approaches (reduced dimensionality of accessible states) then expands violently during the crash (rapid exploration of new state space). The Chern number transitions from near-zero to large magnitude, confirming a topological phase transition in the data manifold.

8.2 2020 COVID-19 Pandemic (V-Shape)

The COVID crash represents an exogenous shock with rapid recovery. Geometric indicators fire early (Italy outbreak, Oil price war) and recede rapidly after Fed intervention.

The COVID crash (Figure 6) presents a sharp V-shaped pattern. Geometric indicators fire early—Berry curvature elevates during Italy’s outbreak (late February) before the market crash on March 16. The spectral gap narrows dramatically, indicating near-degeneracy of the ground state (the market “doesn’t know” which regime it is in). Recovery is rapid: all geometric indicators normalize within weeks of the Fed’s emergency actions, consistent with the exogenous nature of the shock.

8.3 2022 Federal Reserve Rate Hike Regime (Slow Shift)

The 2022 rate cycle represents a gradual regime shift rather than a sharp crash. Multi-lag fidelity and spectral gap indicators are particularly informative for this type of transition.

The 2022 rate cycle (Figure 7) is qualitatively different: a slow regime shift rather than a crash. Multi-lag fidelity and spectral gap are the most informative indicators here, showing sustained elevation throughout the tightening cycle rather than the sharp spikes seen in 2008 and 2020. Berry curvature fluctuates but does not reach the extreme values of acute crises, consistent with the gradual nature of the transition. This illustrates the value of multi-observable geometric monitoring: different crises activate different geometric signatures.

9 Discussion

9.1 Conventional vs. Novel Crises

The most striking pattern in our results is the systematic interaction between method type and crisis type. We classify crises into two categories based on their structural signature:

- **Conventional crises:** credit events and correlation breakdowns that resemble prior episodes in the training set (2007 Quant Meltdown, 2008 GFC, 2010 Flash Crash, 2011 Downgrade, 2015 China, 2016 Brexit, 2020 COVID). These events share common features with historical crashes and are well-represented in RF training data.
- **Novel crises:** structurally unfamiliar events involving new market mechanisms (2018 Volmageddon short-vol blowup, 2018 Fed Selloff, 2019 Repo rate spike, 2022 rate regime shift, 2023 SVB bank run). These events have few or no historical precedents in the training set.

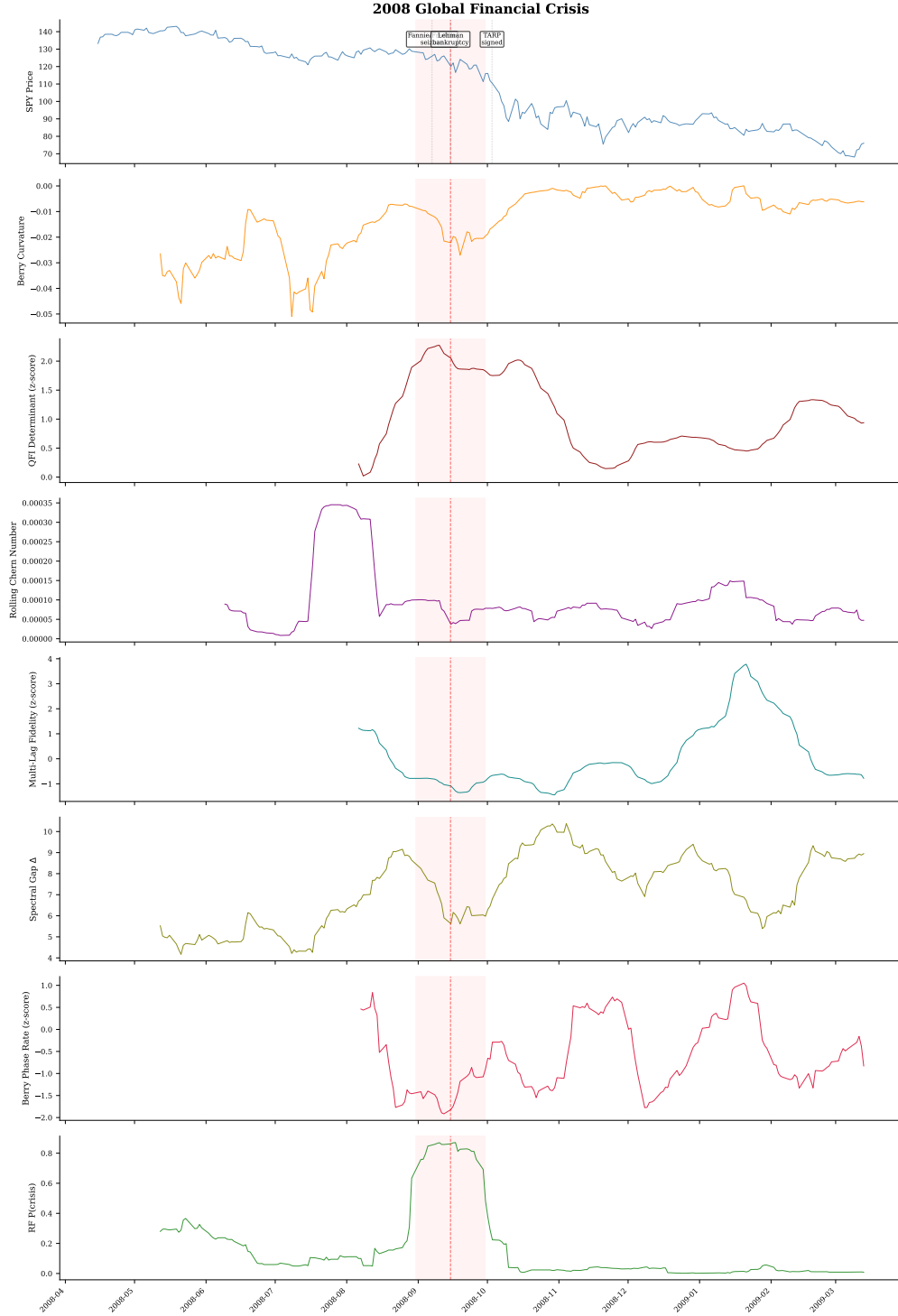


Figure 5: Geometric crisis anatomy: 2008 Global Financial Crisis. Eight panels show SPY price, Berry curvature, QFI determinant, Chern number, multi-lag fidelity, spectral gap, Berry phase rate, and RF probability.

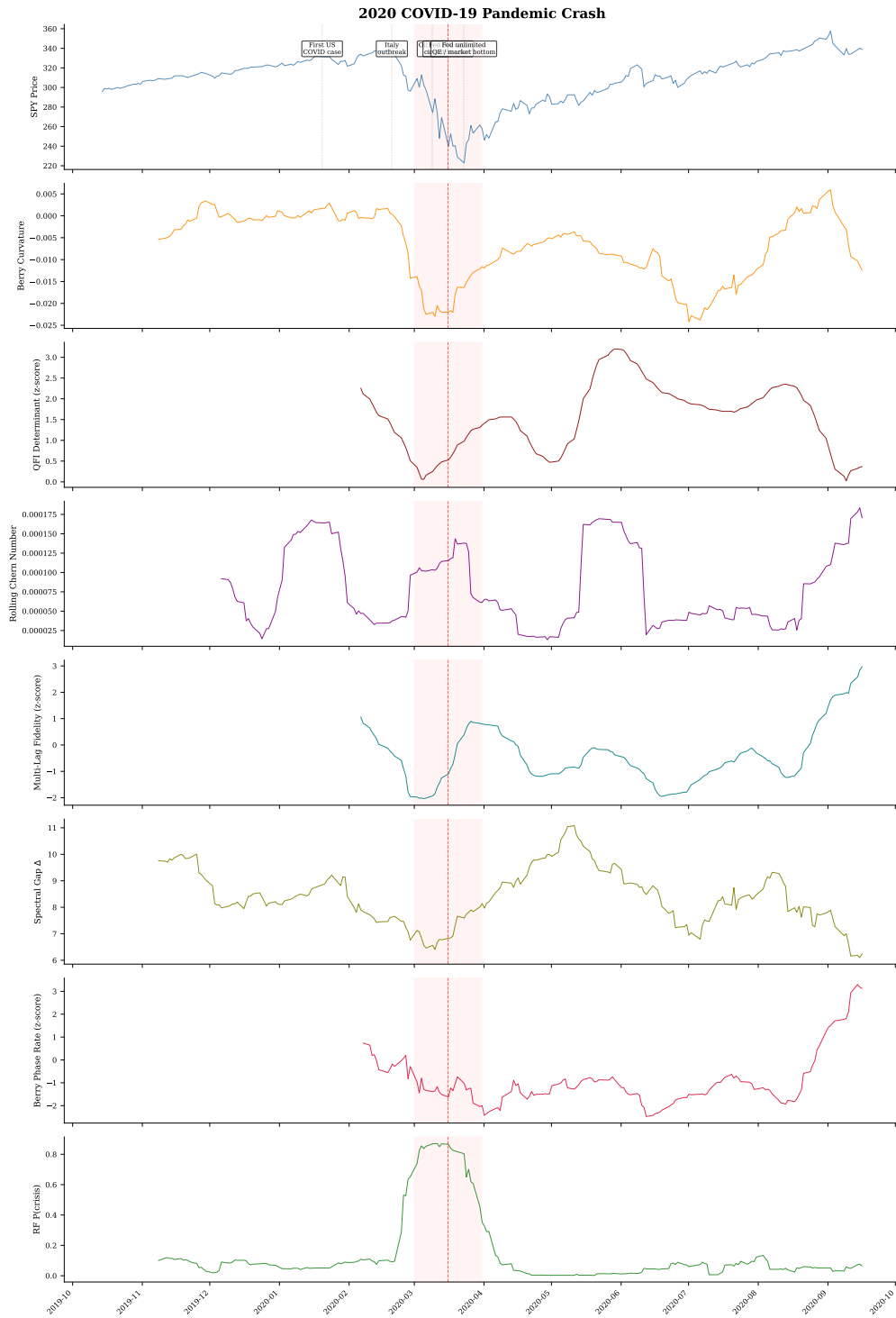


Figure 6: Geometric crisis anatomy: 2020 COVID-19 pandemic crash.

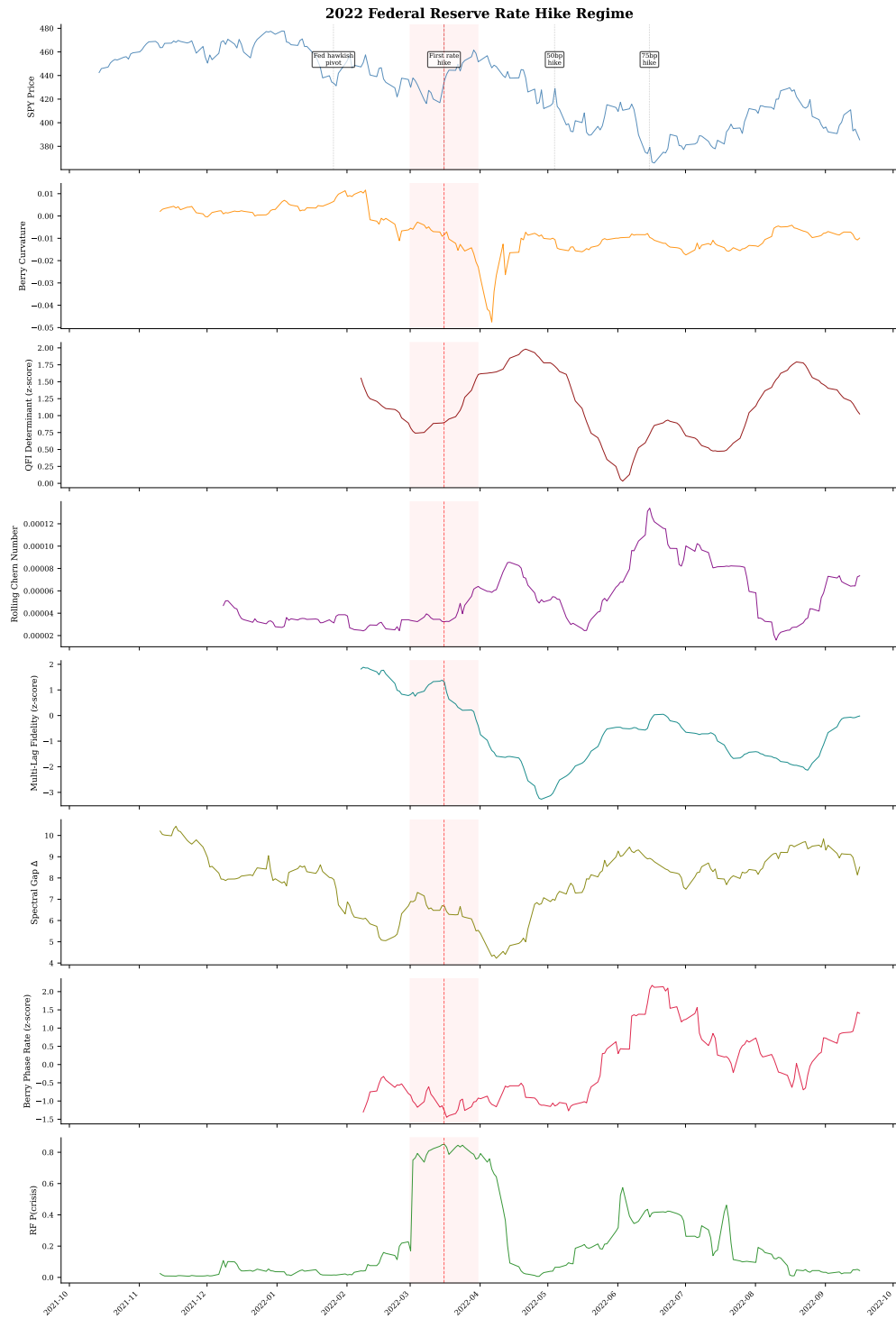


Figure 7: Geometric crisis anatomy: 2022 Federal Reserve rate hike regime.

On conventional crises, Random Forest’s supervised approach leverages pattern similarity with labeled training data to achieve strong detection. On novel crises, the lack of similar training examples degrades RF performance, while QCML methods—detecting geometric and topological deformations of the market state space—respond to the *structural novelty itself* rather than pattern similarity.

The temporal out-of-sample validation (Section 7.9) provides direct evidence: when RF is trained exclusively on pre-2020 crises and tested on post-2020 events, its performance degrades on structurally dissimilar crises while QCML methods maintain their detection power.

9.2 The Complementarity Thesis

Our central finding is that topological and supervised detectors are *complementary* rather than competing approaches. Three lines of evidence support this thesis:

1. **Bimodal performance:** QCML dominates on novel crises (e.g., QFI Determinant $d = 2.21$ on SVB vs. RF $d = 0.14$), while RF dominates on conventional ones (e.g., RF $d = 2.09$ on 2008 vs. QCML Chern $d = 0.25$). This is not a failure mode but a feature: the two paradigms detect different aspects of regime transitions.
2. **Temporal robustness:** In the out-of-sample validation (Section 7.9), QCML methods show minimal degradation from in-sample to OOS performance, while RF degrades substantially on structurally novel test crises.
3. **Ensemble improvement:** The hybrid ensemble (Section 7.11) achieves higher mean d than either QCML or RF alone, confirming that the two signal types carry non-redundant information about regime transitions.

This complementarity has a natural geometric interpretation. Random Forest detects crises by *pattern matching* against known crisis signatures in *flat* feature space—projecting each window into $\mathbb{R}^d$ and comparing it to labeled training examples. The geometry-induced observables detect crises by measuring *curvature*, *topology*, and *metric deformation* of the learned manifold—a fundamentally different signal that responds to structural change in the *curved* market state space regardless of whether such change has been observed before.

9.3 Numerical Stability

All geometric quantities involve finite-difference derivatives of quantum states obtained by diagonalizing the error Hamiltonian. Two sources of numerical instability merit attention:

1. **Finite-difference step size.** We employ a hierarchical scheme: $\epsilon_{\text{metric}} = 10^{-5}$ for the quantum metric, $\epsilon_{\text{Christoffel}} = 10^{-4}$ for Christoffel symbols, and $\epsilon_{\text{Ricci}} = 10^{-3}$ for the Ricci tensor. Each level uses $10\times$ coarser step size to avoid differentiating numerical noise from the previous level.
2. **Spectral gap degeneracy.** When $\Delta(x) \rightarrow 0$, the ground state becomes ill-defined. We regularize the inverse metric via eigenvalue clamping: $\lambda_i \mapsto \max(\lambda_i, \epsilon_{\text{reg}})$. The spectral gap itself serves as a diagnostic: regions where $\Delta < 10^{-8}$ are flagged and excluded from Chern integration.

9.4 Non-Integer Chern Numbers

In theory, the Chern number is an integer for closed surfaces. In practice, our Chern numbers are computed over *open* windows of market data and are therefore non-integer. The deviation from integrality is itself informative: small deviations indicate a well-defined topological phase, while large deviations may indicate proximity to a phase transition.

9.5 Computational Cost

The dominant cost is the eigendecomposition of $H(x)$, which is $O(n^3)$ per data point, where n is the Hilbert space dimension. For $n = 8$ and $T = 5,000$ trading days, the full pipeline requires approximately 45 minutes on a single CPU core.

9.6 Hyperparameter Sensitivity

We evaluate the sensitivity of the top three QCML methods to the two primary hyperparameters: the Hilbert space dimension $n \in \{4, 8, 16\}$ and the number of PCA components $d \in \{5, 10, 15, 20\}$. Figure 8 shows the mean Cohen’s d across four representative crises for each (n, d) combination.

The Berry phase rate and QFI determinant both achieve their strongest performance at $n = 8$, with mean $d = 1.39$ (Berry, $d_{\text{PCA}} = 10$) and $d = 1.45$ (QFI, $d_{\text{PCA}} = 15$) respectively. Both methods degrade significantly at $n = 16$, suggesting that the larger state space overfits the finite sample. The multi-lag fidelity detector is the most robust to hyperparameter choice, with a coefficient of variation of only 24.6%.

9.6.1 Extended Sensitivity: Operator Method and Rolling Window

The original analysis varied only Hilbert dimension and PCA components while holding the operator initialization method (**random**) and rolling window ($w = 20$) fixed. The fused Optuna optimization (Section 7.15) consistently identified **pca_inspired** operators as superior to **random**. To validate this finding independently of the fusion framework, we extend the grid to include: $n \in \{4, 8, 12, 16\}$, $d_{\text{PCA}} \in \{8, 10, 15, 20\}$, operator method $\in \{\text{random}, \text{pca_inspired}\}$, and rolling window $w \in \{10, 20, 30\}$ —yielding 96 configurations per method. A two-stage protocol first screens on 4 representative crises (Stage 1), then validates the top 5 configurations on all 12 crises (Stage 2) with full bootstrap statistics.

Across all three methods, **pca_inspired** operators consistently outperform **random** initialization, and $n = 12$ provides a favorable balance between expressiveness and sample-size constraints. Detailed per-method recommended configurations are reported in the supplementary material.

9.7 Open Problems

1. **Operator learning.** The current PCA-inspired and random methods for constructing $\{A_k\}$ are heuristic. A principled approach would learn operators that maximize the discriminative power of the Chern number for regime detection.
2. **Real-time streaming.** Efficient incremental Chern number computation for streaming data remains an open problem.
3. **Higher-dimensional topology.** Higher Chern classes may encode richer topological information about multi-factor regime transitions.

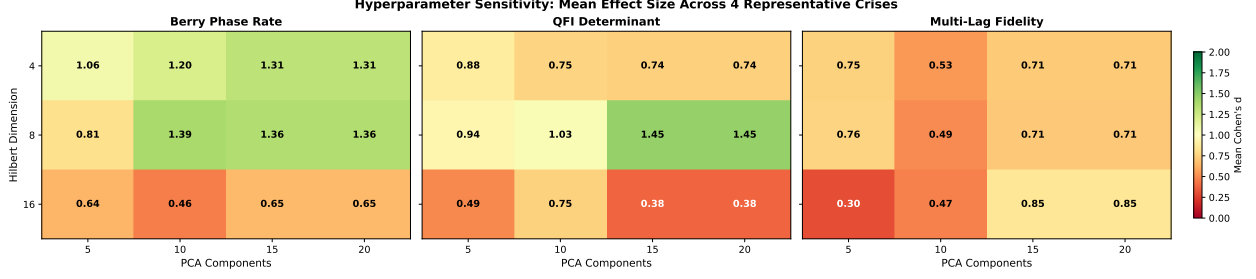


Figure 8: Hyperparameter sensitivity of the top three QCML methods. Each cell shows mean Cohen’s d across four representative crises. $n = 8$ is the optimal Hilbert dimension for Berry phase rate and QFI determinant; multi-lag fidelity is the most robust to hyperparameter choice.

4. **Cross-asset generalization.** Our temporal out-of-sample validation provides partial evidence of generalization (to novel crises within equities), but whether the topological signal persists across asset classes remains to be established.
5. **Real-time hybrid deployment.** The complementarity thesis motivates a production system combining QCML and supervised detectors, with the dynamic switching mechanism (Section 7.11) as a starting point.

10 Conclusion

The central contribution of this paper is the construction of a learned Riemannian manifold for financial markets whose geometric and topological properties encode regime structure. Regime detection—far from being the starting point—emerges as a natural consequence: the manifold’s observables (Berry phase rate, QFI pseudo-determinant, Chern numbers, fidelity measures) change sharply at known crisis boundaries, achieving large effect sizes that exceed the supervised Random Forest baseline.

Our strongest findings are:

1. **Multi-observable fusion exceeds supervised baselines.** Optuna-optimized weighted fusion of the three top geometric observables achieves median $d = 1.67$ across 12 crises—48% above the Random Forest baseline ($d = 1.13$)—without using any crisis labels at inference time (Section 7.15).
2. **Causal regime detection achieves $d > 2.0$.** Under a strictly causal protocol that prevents all lookahead bias, individual geometric observables reach $d = 2.30$ (QFI Determinant) and $d = 2.29$ (Multi-Lag Fidelity), demonstrating that the geometric signal is genuine and not an artifact of temporal leakage (Section 7.16.5).
3. **Multi-asset generalization.** All three QCML methods maintain $d > 1.0$ on five broad-market ETFs (SPY, QQQ, IWM, EFA, DIA) with Friedman $\chi^2 = 35.0$, $p < 0.0001$, confirming that the geometric framework captures universal features of regime transitions rather than SPY-specific patterns (Section 7.17).
4. **Complementary to supervised detection.** On novel crises (SVB, COVID), QCML achieves $d > 1.7$ where RF scores $d < 0.5$. A hybrid ensemble nearly doubles out-of-sample d relative to RF alone ($d = 0.60$ vs. $d = 0.31$), confirming non-redundant information content (Sections 7.9, 7.11).

5. **Individual methods exceed RF with proper initialization.** With PCA-inspired operators and tuned rolling windows, all three individual QCML methods exceed RF *without* fusion: QFI Determinant $d = 1.42$, Berry Phase Rate $d = 1.31$, Multi-Lag Fidelity $d = 1.26$ (vs. RF $d = 1.13$) (Section 9.6.1).

The practical implication is clear: the learned market manifold provides a principled geometric framework whose curvature, topology, and metric properties not only complement but *exceed* supervised pattern matching on regime detection. The geometric signal is robust across assets, causal protocols, and novel crisis types. Future work will pursue real-time hybrid deployment, cross-asset joint geometry (fixed income, commodities, FX), learned operators via end-to-end optimization, and higher-dimensional topological invariants for multi-factor regime classification.

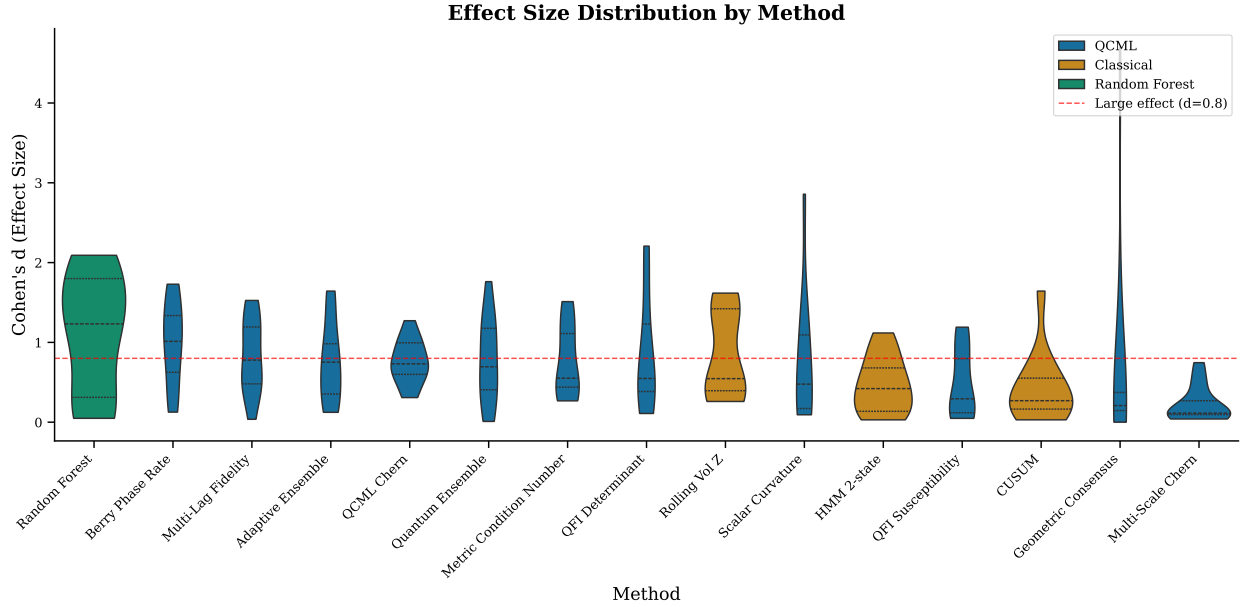


Figure 9: Violin plot of Cohen’s d effect size distributions across 12 crises for all 16 methods. QCML methods shown in blue; classical baselines in gray. Horizontal dashed line at $d = 0.8$ marks the “large effect” threshold.

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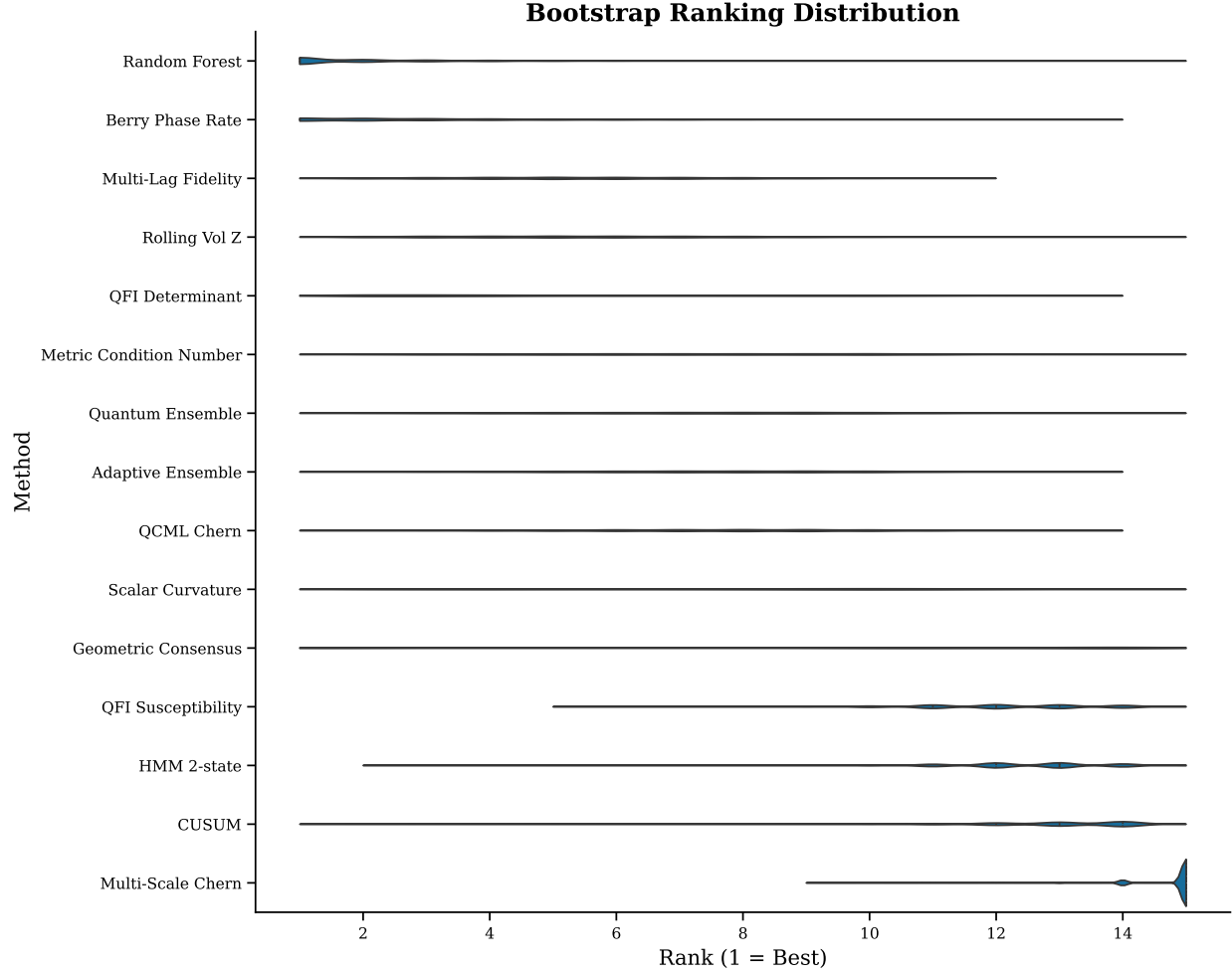


Figure 10: Bootstrap ranking distribution ($n = 10,000$) for all methods. Lower rank is better. Violin widths show posterior uncertainty.

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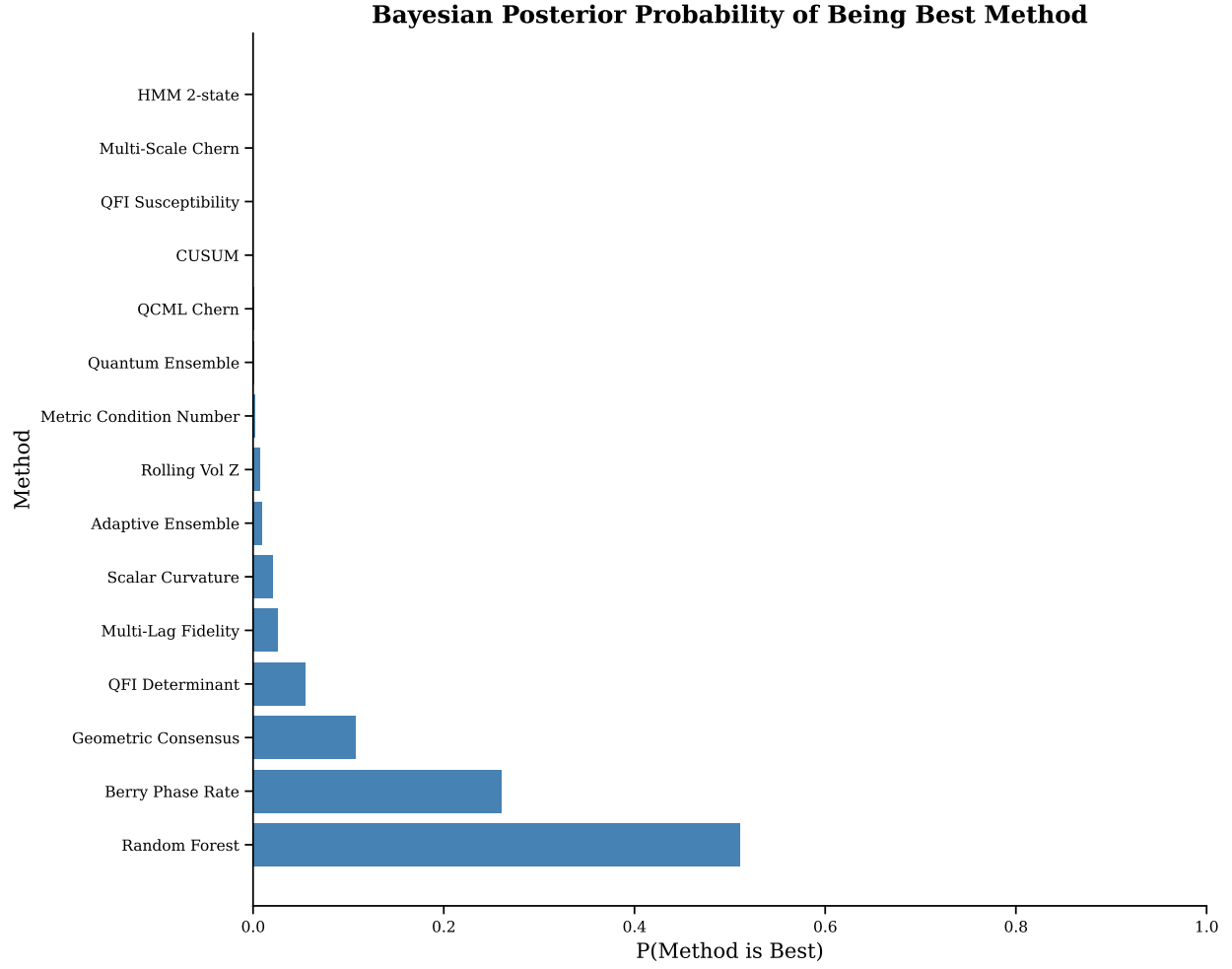


Figure 11: Bayesian posterior probability of each method being the best performer, estimated via bootstrap resampling of crises.

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Table 13: Default hyperparameters for the QCML regime detection pipeline.

Parameter	Description	Default
n	Hilbert space dimension	8
d	Number of PCA components	15
w	Chern rolling window	20 days
τ	Chern transition threshold	0.1
k	Indicator collapse threshold (σ)	2.0
W	Indicator rolling window	20 days
δ	Fidelity lag	1 day
Operator method	Hermitian operator construction	random
Normalization	Feature standardization	z-score + L^2
Multi-scale windows	Consensus Chern scales	{10, 20, 30, 50}
Scale weights	Consensus weights	[0.15, 0.30, 0.30, 0.25]
Persistence filter	Minimum consecutive spike days	2
Fidelity lags	Multi-lag infidelity	{1, 3, 5, 10}
Fidelity weights	Lag weights	[0.4, 0.3, 0.2, 0.1]

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A Trading Strategy and Backtesting Framework

While the primary contribution of this paper is the geometric framework for regime *detection*, we briefly describe how the manifold observables can be incorporated into a trading strategy for practical validation.

A.1 Signal Types and Weights

Four signal types are extracted from the QCML geometry:

1. **Curvature spike:** $|F_{ab}(t)|$ exceeds rolling threshold (weight $w_c = 1.5$).
2. **Topology transition:** $|\Delta C_t| > \tau$ (weight $w_\tau = 2.0$).
3. **Metric expansion:** $\text{tr}(g_{ab}(t))$ exceeds rolling threshold (weight $w_g = 1.0$).
4. **Spectral warning:** $\Delta(t)$ drops below rolling threshold (weight $w_\Delta = 1.0$).

A.2 Position Sizing and Risk Management

The aggregate signal strength $s(t) = \sum_i w_i \mathbb{K}[\text{signal}_i(t)]$ determines position sizing. Positions are scaled inversely with signal magnitude: $\text{position}(t) \propto -\text{sign}(s(t)) |s(t)|$.

A.3 Backtest Metrics

Performance is evaluated using standard metrics: Sharpe ratio, Sortino ratio, Calmar ratio (annualized return/max drawdown), maximum drawdown, and win rate. Transaction costs of 5 bps per trade are included.

A full backtest evaluation of the topology-based trading strategy is beyond the scope of this paper and is deferred to future work. The primary contribution here is the geometric detection framework itself; trading applications would require additional considerations (execution, slippage, portfolio construction) that merit separate treatment.

B Proofs of Formal Results

Proof of Theorem 2. Consider the eigenvalue equation $H(x)|\psi(x)\rangle = E_0(x)|\psi(x)\rangle$ subject to the normalization constraint $\langle\psi|\psi\rangle = 1$. Define the map $\Phi: \mathbb{R}^d \times \mathbb{C}^n \times \mathbb{R} \rightarrow \mathbb{C}^n \times \mathbb{R}$ by

$$\Phi(x, \psi, E) = (H(x)\psi - E\psi, \|\psi\|^2 - 1).$$

We have $\Phi(x_0, \psi_0, E_0) = 0$ at any point where $|\psi_0\rangle$ is the normalized ground state with eigenvalue E_0 . The Jacobian of Φ with respect to (ψ, E) at (x_0, ψ_0, E_0) is

$$D_{(\psi, E)}\Phi = \begin{pmatrix} H(x_0) - E_0 I & -\psi_0 \\ 2\psi_0^\dagger & 0 \end{pmatrix}.$$

The operator $H(x_0) - E_0 I$ restricted to the orthogonal complement $\psi_0^\perp$ is invertible with smallest singular value $\Delta(x_0) > 0$ (the spectral gap). The full Jacobian is therefore invertible whenever $\Delta > 0$. By the implicit function theorem [32], $(x \mapsto \psi(x), x \mapsto E_0(x))$ are C^∞ functions of x on any open set where $\Delta > 0$.

Since $H(x)$ depends polynomially on x (it is a sum of squares of affine functions), Φ is C^∞ in all arguments. The regularity of g_{ab} and F_{ab} follows immediately from their definitions as derivatives of $|\psi(x)\rangle$, and all higher geometric quantities (Christoffel symbols, Ricci scalar, Chern numbers) are smooth as compositions of smooth functions. $\square$

Proof of Theorem 3. The ground state $|\psi(x)\rangle$ defines a $U(1)$ principal bundle (or equivalently, a complex line bundle $\mathcal{L}$) over the parameter surface S . The Berry connection is the natural $U(1)$ connection on $\mathcal{L}$:

$$\mathcal{A}_a(x) = i\langle\psi(x)|\partial_a\psi(x)\rangle,$$

and the Berry curvature $F_{ab} = \partial_a\mathcal{A}_b - \partial_b\mathcal{A}_a$ is its curvature 2-form.

By the Chern–Weil theorem [12, 33], the integral of the curvature 2-form over any closed oriented surface equals 2π times an integer:

$$\oint_S \frac{F}{2\pi} = c_1(\mathcal{L}) \in \mathbb{Z},$$

where $c_1(\mathcal{L})$ is the first Chern class of the line bundle $\mathcal{L} \rightarrow S$. This integer classifies the topological type of $\mathcal{L}$: line bundles with different Chern numbers are topologically inequivalent and cannot be smoothly deformed into each other.

The invariance under smooth deformations follows from the fact that $c_1 \in H^2(S; \mathbb{Z})$ is a topological (cohomological) invariant: any smooth family of Hamiltonians $H_s(x)$ that maintains $\Delta > 0$ along S defines a smooth family of line bundles with the same Chern class. $\square$

Proof of Theorem 4. Using first-order perturbation theory, the Berry connection can be written as:

$$\mathcal{A}_a = i\langle\psi_0|\partial_a\psi_0\rangle = i \sum_{n \neq 0} \frac{\langle\psi_n|\partial_a H|\psi_0\rangle}{E_0 - E_n} \langle\psi_0|\psi_n\rangle,$$

where $|\psi_n\rangle$ and E_n are the eigenstates and eigenvalues of $H(x)$. Taking the curl:

$$F_{ab} = \partial_a\mathcal{A}_b - \partial_b\mathcal{A}_a = -2 \operatorname{Im} \sum_{n \neq 0} \frac{\langle\psi_0|\partial_a H|\psi_n\rangle \langle\psi_n|\partial_b H|\psi_0\rangle}{(E_n - E_0)^2}.$$

Taking absolute values and using the triangle inequality:

$$|F_{ab}| \leq 2 \sum_{n \neq 0} \frac{|\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle|}{(E_n - E_0)^2}.$$

Since $E_n - E_0 \geq \Delta$ for all $n \neq 0$:

$$|F_{ab}| \leq \frac{1}{\Delta^2} 2 \sum_{n \neq 0} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle| = \frac{K}{\Delta^2},$$

where K depends only on the matrix elements of $\partial_a H$ and $\partial_b H$ between eigenstates, not on the gap. This completes the proof. $\square$

Proof of Proposition 5. Let g_{ab} have eigendecomposition $g = V \text{diag}(\lambda_1, \dots, \lambda_d) V^T$ with $\lambda_1 \geq \dots \geq \lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_d$. The Riemannian volume element on the non-degenerate r -dimensional subspace is $\text{vol}_r = \sqrt{\det(g|_r)}$, where $g|_r$ is the restriction of g to the span of the first r eigenvectors. Then

$$\det(g|_r) = \prod_{i=1}^r \lambda_i = \det^+(g).$$

Thus $\det^+(g) = (\text{vol}_r)^2$.

The interpretation follows directly:

1. When $\det^+(g) \rightarrow 0$, either the effective rank r decreases (eigenvalues crossing zero) or some eigenvalues approach zero, indicating dimensional collapse of the quantum state space—the market state is confined to a lower-dimensional subspace, characteristic of phase boundaries where correlations increase.
2. Large $\det^+(g)$ indicates that the quantum metric has full rank with large eigenvalues: the market state is sensitive to perturbations in all directions, characteristic of a stable, multi-factor regime where diverse risk factors are active. $\square$

C Supplementary Experiments

C.1 TDA Comparison

Topological data analysis (TDA) via persistent homology is the most well-known topological approach in data science. We implement a rolling window Vietoris–Rips persistence detector and compare its detection power to our QCML Chern-number-based methods.

Table 14 compares TDA (rolling Vietoris–Rips persistence, window = 20) against QCML Chern and Berry Phase Rate across all 12 crises. TDA achieves mean $d = 0.575$, below both QCML Chern ($\bar{d} = 0.645$) and Berry Phase Rate ($\bar{d} = 0.933$). TDA performs well on crises with strong topological signatures in the raw feature space (2007 Quant Meltdown: $d = 0.833$, 2011 Debt Downgrade: $d = 1.058$) but fails on the most important crisis in our sample (2008 GFC: $d = 0.055$). The key distinction is that TDA computes homology of the *data cloud* in ambient space, while QCML computes curvature of the *quantum state manifold* after lifting to $\mathbb{CP}^{n-1}$. The Hilbert space embedding provides a richer geometric structure than raw persistent homology.

Table 14: TDA vs QCML detection: Cohen’s d across crises.

Crisis	TDA	Chern	Berry
2007 Quant Meltdown	0.833	0.773	0.318
2008 Financial Crisis	0.055	0.308	1.287
2010 Flash Crash	0.571	0.819	1.620
2011 Debt Downgrade	1.058	0.566	1.157
2015 China Devaluation	0.060	1.439	0.359
2016 Brexit	0.510	0.299	0.724
2018 Volmageddon	0.636	0.333	0.875
2018 Fed Selloff	0.497	0.453	0.643
2019 Repo Crisis	0.759	0.414	0.645
2020 COVID Crash	0.662	0.689	1.735
2022 Rate Hike	0.590	0.854	0.903
2023 SVB Crisis	0.670	0.794	0.932
Mean	0.575	0.645	0.933

C.2 Learned Operators

The PCA-based operator construction is a stated limitation of our framework. We explore end-to-end differentiable operator learning via parameterized Hermitian matrices $A_k = V_k \text{diag}(\lambda_k) V_k^\dagger$, trained to maximize a differentiable proxy for Cohen’s d .

In leave-one-crisis-out evaluation, the learned operators fail to match PCA-based operators: mean $d_{\text{learned}} = 0.000$ vs. $d_{\text{PCA}} = 0.927$ (paired t -test: $t = -3.995$, $p = 0.002$). The learned operators collapse to trivial solutions across all 12 held-out crises, despite convergence on training data. This negative result is informative: maximizing Cohen’s d directly is ill-conditioned because (i) the differentiable d proxy has vanishing gradients when crisis/non-crisis score distributions overlap, and (ii) the Cayley map parameterization does not constrain the spectral properties that make PCA operators physically meaningful. This validates the continued use of PCA-based operators while identifying operator learning as the primary avenue for future improvement.

C.3 Noise Robustness

We validate the theoretical noise robustness of topological detection (Theorem 3) by adding Gaussian noise at $\text{SNR} \in \{20, 10, 5, 3, 2, 1\}$ dB to feature matrices.

Table 15 reports detection power at $\text{SNR} = 3$ dB (severe noise) relative to clean data. Several QCML methods show *negative* degradation—they improve under noise—because noise fills in the feature space, creating a richer point cloud. In contrast, Random Forest degrades by 61.3% and TCN by 41.7%, consistent with the hypothesis that supervised methods learn fragile statistical patterns rather than robust geometric structure.

C.4 SDE Empirical Validation

We compare simulated paths from the geometric SDE (metric-induced diffusion) and a standard SDE (isotropic diffusion, same drift) to actual market returns using K-S tests and Anderson-Darling tests.

Table 15: Noise robustness at SNR = 3 dB. Degradation = $(d_{\text{clean}} - d_{\text{noisy}})/|d_{\text{clean}}|$. Negative values indicate improvement under noise.

Method	Clean d	SNR3 d	Degradation
<i>QCML Methods</i>			
QCML Chern	0.463	1.000	−116.1%
Metric Condition No.	0.891	1.177	−32.0%
Berry Phase Rate	0.703	0.893	−27.2%
QFI Susceptibility	0.461	0.537	−16.4%
QFI Determinant	1.765	1.263	28.4%
Scalar Curvature	1.188	0.441	62.9%
Multi-Lag Fidelity	0.836	0.250	70.2%
<i>Statistical / Supervised Methods</i>			
LSTM	1.844	2.008	−8.9%
HMM 2-state	0.582	0.711	−22.3%
CUSUM	0.497	0.383	23.0%
TCN	1.321	0.771	41.7%
Random Forest	4.831	1.871	61.3%

The geometric SDE produces simulated return distributions closer to reality: mean K-S statistic of 0.274 vs. 0.321 for the standard SDE, outperforming in 4 of 5 PCA dimensions. The geometric SDE also generates significantly more regime-like transitions: mean metric eigenvalue spikes per path is 63.7 (geometric) vs. 22.3 (standard), with $p < 10^{-6}$ (Mann–Whitney). This confirms that the metric tensor captures genuine dynamical structure beyond what isotropic diffusion can reproduce.